SignNet-II: An End-to-End Transformer Architecture for Bidirectional Sign Language Translation

Abstract:

Sign language is a vital communication medium for the Deaf and hard-of-hearing communities. However, the gap between sign language users and non-signers often leads to communication barriers. SIGNNET II proposes a novel transformer-based architecture designed for two-way translation between sign language and spoken language, facilitating seamless interaction. Leveraging the strengths of transformer models in capturing long-range dependencies, SIGNNET II effectively models the complex spatial and temporal dynamics inherent in sign language gestures.

The model employs a dual-stream input system to process both video sequences of sign language and textual data, enabling bidirectional translation capabilities. By incorporating advanced attention mechanisms, SIGNNET II accurately aligns sign language gestures with corresponding spoken language tokens, improving translation accuracy and fluency. This two-way approach not only translates sign language into text but also generates sign language sequences from textual input, making it a comprehensive communication tool.

To evaluate SIGNNET II, extensive experiments were conducted on publicly available sign language datasets, including large-scale benchmarks with diverse vocabularies and signer variations. The model demonstrated significant improvements in translation quality, outperforming existing state-of-the-art methods in terms of BLEU scores for text generation and accuracy metrics for sign recognition. Additionally, SIGNNET II showed robustness in handling continuous signing with complex sentence structures.

Beyond technical performance, SIGNNET II emphasizes practical applicability by integrating real-time inference capabilities and user-friendly deployment options. This makes it suitable for live communication scenarios such as video conferencing and assistive technologies. The model's architecture is also designed to be extensible, allowing adaptation to multiple sign languages and dialects through transfer learning.

In conclusion, SIGNNET II represents a significant advancement in sign language translation technology, bridging communication gaps with a scalable, accurate, and bidirectional transformer-based model. Its success paves the way for more inclusive communication systems, enhancing accessibility for the Deaf community worldwide.

Introduction:

Sign language serves as the primary means of communication for millions of Deaf and hard-of-hearing individuals worldwide. Despite its rich linguistic structure and cultural significance, sign language remains largely inaccessible to non-signers, leading to significant communication barriers. Bridging this gap through effective translation between sign language and spoken or written language is crucial for fostering inclusivity and enabling seamless interaction in diverse social, educational, and professional contexts.

Traditional approaches to sign language translation have often relied on handcrafted features or sequential models such as recurrent neural networks (RNNs) and long short-term memory (LSTM) networks. While these methods have achieved notable progress, they face challenges in modeling the complex spatial-temporal dependencies present in sign language, which involves intricate hand gestures, facial expressions, and body movements. Moreover, many existing models focus primarily

on one-way translation, either from sign to text or text to sign, limiting their practical usability in real-world communication.

Recent advances in transformer architectures, which excel at capturing long-range dependencies and contextual relationships, offer promising opportunities for improving sign language translation. Transformers' self-attention mechanisms allow models to effectively understand and align multimodal input sequences, making them well-suited for the bidirectional nature of sign language translation. Leveraging this, SIGNNET II introduces a transformer-based two-way translation framework that simultaneously addresses sign-to-text and text-to-sign translation within a unified model.

This work presents the architecture, training methodology, and evaluation of SIGNNET II, highlighting its improvements over existing models in terms of translation accuracy, robustness, and scalability. By integrating advanced attention mechanisms and dual-stream input processing, SIGNNET II captures the rich multimodal features of sign language and achieves fluent, context-aware translations. The model's design also supports real-time inference, making it applicable to practical scenarios such as live interpretation and assistive communication technologies.

Ultimately, SIGNNET II aims to reduce communication barriers and empower the Deaf community by providing a comprehensive, accurate, and scalable sign language translation solution. The contributions of this work not only advance the state-of-the-art in sign language processing but also lay the foundation for future research and applications in inclusive communication technologies.

Literature Survey:

1. **"Sign Language Recognition Using Deep Learning"**
   Authors: Koller et al. (2019)
   Description: Introduced a convolutional neural network (CNN) and recurrent neural network (RNN) hybrid approach for recognizing isolated signs and continuous sign language from video sequences. Highlighted challenges of temporal modeling and spatial feature extraction in sign language recognition.

2. **"End-to-End Sign Language Translation with Transformers"**
   Authors: Camgoz et al. (2020)
   Description: Proposed a transformer-based architecture for sign language translation that directly converts sign videos into spoken language sentences. Demonstrated the transformer's superiority over RNNs in capturing long-term dependencies in continuous signing.

3. **"Multi-Modal Sign Language Translation via Attention-Based Neural Networks"**
   Authors: Huang et al. (2021)
   Description: Developed an attention-based multi-modal framework combining video, hand pose, and facial expression data to improve translation accuracy. Emphasized the importance of multi-modal cues in sign language understanding.

4. **"Two-Way Neural Machine Translation for Sign Language"**
   Authors: Zhang & Li (2022)
   Description: Presented a bidirectional sign language translation system capable of translating text to sign glosses and glosses to text. Used sequence-

to-sequence models with attention, highlighting the need for two-way translation in practical applications.

5. **"Transformer-Based Approaches for Sign Language Recognition and Translation"**
   Authors: Singh et al. (2023)
   Description: Surveyed recent transformer applications in sign language tasks, discussing architectural adaptations for spatial-temporal features and challenges like data scarcity. Proposed hybrid transformer designs combining CNN encoders with transformer decoders.

Existing system:

Sign language translation has traditionally been approached through isolated sign recognition, where individual gestures are detected and classified using handcrafted features or classical machine learning techniques. Early systems relied heavily on computer vision methods such as skin-color segmentation, motion tracking, and feature engineering to identify static or dynamic signs. These methods, while pioneering, struggled with scalability and generalization, especially in continuous signing scenarios with varied backgrounds and signer styles.

With the advent of deep learning, many existing systems shifted towards leveraging convolutional neural networks (CNNs) and recurrent neural networks (RNNs) to extract spatiotemporal features from video sequences. These architectures enabled improved modeling of the temporal dynamics in sign language. Notable systems used CNNs for frame-level feature extraction combined with long short-term memory (LSTM) networks to capture sequential dependencies. However, such models often suffered from limitations like vanishing gradients and difficulty in

capturing long-range contextual information, which is crucial for accurate translation.

More recently, transformer-based models have gained traction due to their ability to handle long-range dependencies using self-attention mechanisms. Existing transformer systems in sign language translation primarily focus on one-way translation—either converting sign videos into spoken language text or generating sign glosses from textual input. For instance, some models translate sign videos into textual sentences with high accuracy by aligning video features and language tokens. Despite these advancements, most systems lack the capability to perform two-way translation within a unified framework, limiting their usability in real-time conversational contexts.

In terms of datasets and evaluation, existing systems generally rely on benchmark datasets like RWTH-PHOENIX-Weather, CSL, and ASLLVD, which provide annotated sign language videos and corresponding gloss or text translations. While these datasets have driven research forward, many models exhibit reduced performance when faced with diverse signers, varying signing speeds, or complex sentence structures. Moreover, real-time applicability remains a challenge due to computational complexity and latency issues in current architectures.

Overall, while the progress in sign language translation models is significant, gaps remain in achieving robust, bidirectional, and real-time translation capabilities. Existing systems mostly focus on either sign-to-text or text-to-sign translation but rarely both, and often do not fully exploit multimodal inputs like hand pose and facial expressions. SIGNNET II aims to address these limitations by providing a transformer-based two-way translation model that integrates multimodal features and supports practical deployment scenarios.

Disadvantages of Existing Systems:

1. **Limited Bidirectional Translation**

   Most existing sign language translation models focus on one-way translation—either from sign language to text or from text to sign language. This limitation restricts real-time, two-way communication, which is essential for natural conversations between signers and non-signers.

2. **Inadequate Modeling of Multimodal Features**

   Many systems primarily rely on visual cues from video frames without fully integrating other important modalities such as hand pose estimation, facial expressions, and body posture. This leads to incomplete representation of the rich semantic and grammatical information inherent in sign language.

3. **Challenges with Long-Range Dependencies**

   While transformers address some sequence modeling issues, earlier RNN and CNN-based systems struggle with capturing long-term dependencies in continuous sign language, which affects translation fluency and context understanding. Even some transformer models do not adequately capture temporal dynamics in very long sequences.

4. **Performance Drop in Real-World Settings**

   Most existing systems are trained and tested on limited benchmark datasets, which often lack signer diversity, varied backgrounds, and natural signing speeds. Consequently, these models suffer from performance degradation when deployed in real-world scenarios with more variability and noise.

5. **High Computational Complexity and Latency**

   Transformer-based models, though powerful, require significant computational resources for training and inference. This complexity results in

higher latency, making many existing systems unsuitable for real-time applications like live interpretation or mobile assistive devices.

6. **Insufficient Dataset Availability and Scalability**
   Sign language datasets are limited in size, language diversity, and annotation quality. Existing systems often struggle to generalize well due to data scarcity, making it difficult to scale the models across different sign languages or dialects.

Proposed System:

The proposed system, **SIGNNET II**, is a novel transformer-based architecture designed to enable two-way translation between sign language and spoken language, addressing the limitations of existing models. SIGNNET II integrates advanced self-attention mechanisms to effectively capture both spatial and temporal dependencies within continuous sign language video sequences while also managing the complexities of natural language text, making it a comprehensive communication framework.

Unlike previous one-way systems, SIGNNET II features a unified model that performs **bidirectional translation**—translating sign language videos into text and generating sign language sequences from textual input. This two-way capability facilitates seamless, real-time communication between Deaf and hearing individuals, supporting conversational fluency in both directions. The model's architecture is designed to share learned representations, enhancing translation consistency and reducing the need for separate training pipelines.

To capture the rich multimodal characteristics of sign language, SIGNNET II processes multiple input streams, including video frames, hand keypoints, and facial expression features. This multimodal fusion enriches the model's understanding of the linguistic nuances embedded in gestures and expressions, improving the accuracy and naturalness of both recognition and generation tasks. The transformer's attention layers dynamically weigh these inputs to focus on the most informative cues during translation.

SIGNNET II also incorporates a scalable training strategy leveraging large annotated datasets and transfer learning techniques to adapt across different sign languages and dialects. The model employs data augmentation and domain adaptation methods to enhance robustness against signer variability, background noise, and diverse environmental conditions, making it suitable for deployment in real-world scenarios.

Finally, SIGNNET II is optimized for efficient inference to support **real-time translation**, enabling practical use cases such as live video conferencing, assistive communication devices, and educational tools. Its modular design allows easy integration with hardware accelerators and mobile platforms, paving the way for accessible and inclusive communication technology.

System Analysis:

System analysis is crucial to understanding the requirements, constraints, and functional flow of SIGNNET II. It helps break down the model's architecture, its data flow, and its operational characteristics, laying the groundwork for effective implementation and optimization.

1. Problem Definition

The primary challenge addressed by SIGNNET II is the lack of a robust, accurate, and real-time two-way translation system between sign language and spoken/written language. Existing solutions are limited by one-way communication, poor handling of continuous signing, lack of multimodal integration, and high latency. SIGNNET II aims to bridge this gap by offering a unified, transformer-based solution capable of handling complex sign language translation tasks in both directions.

2. Feasibility Study

- **Technical Feasibility:** The use of transformer architecture, which has already demonstrated success in natural language processing and video understanding, ensures technical feasibility. Existing hardware (e.g., GPUs, TPUs) and software libraries (PyTorch, TensorFlow) support the required computations.
- **Operational Feasibility:** SIGNNET II is designed with real-time performance in mind, ensuring it can be deployed in practical scenarios such as live communication, education, and public services.
- **Economic Feasibility:** While initial model training requires substantial computing resources, deployment on edge devices or cloud services can be cost-effective due to optimized inference and transfer learning techniques.

3. Functional Analysis

- **Input Processing:** The system accepts either sign language video input or textual input. For video, preprocessing includes frame extraction, hand keypoint detection, and facial expression analysis.
- **Model Core (Transformer):** The encoder-decoder transformer architecture handles bidirectional translation. For sign-to-text, the encoder captures visual

and temporal features, while the decoder generates corresponding text. For text-to-sign, the process is reversed, generating gesture representations from textual input.

- **Output Generation:** The final output is either a grammatically correct sentence (from signs) or a synthesized sequence of gestures or gloss representations (from text).

4. Non-Functional Requirements

- **Accuracy:** High accuracy in sign recognition and sentence generation is critical. This is achieved through attention mechanisms and multimodal fusion.
- **Scalability:** The architecture is designed to support additional sign languages with minimal retraining.
- **Real-time Performance:** Low latency is prioritized, especially during inference, to make the system usable in live communication scenarios.
- **Robustness:** The system must handle varied signing styles, lighting conditions, and backgrounds effectively.

5. Risk Analysis

- **Data Limitations:** Availability of large, annotated sign language datasets can be a bottleneck. This is mitigated using data augmentation and synthetic data generation.
- **Complexity of Real-Time Inference:** Transformers can be computationally intensive. This is addressed through model compression, pruning, and optimized inference pipelines.

- **Generalization Across Signers:** Variability among signers (speed, style, region) may affect performance. Transfer learning and signer adaptation techniques are used to improve generalization.

SYSTEM REQUIREMENTS:

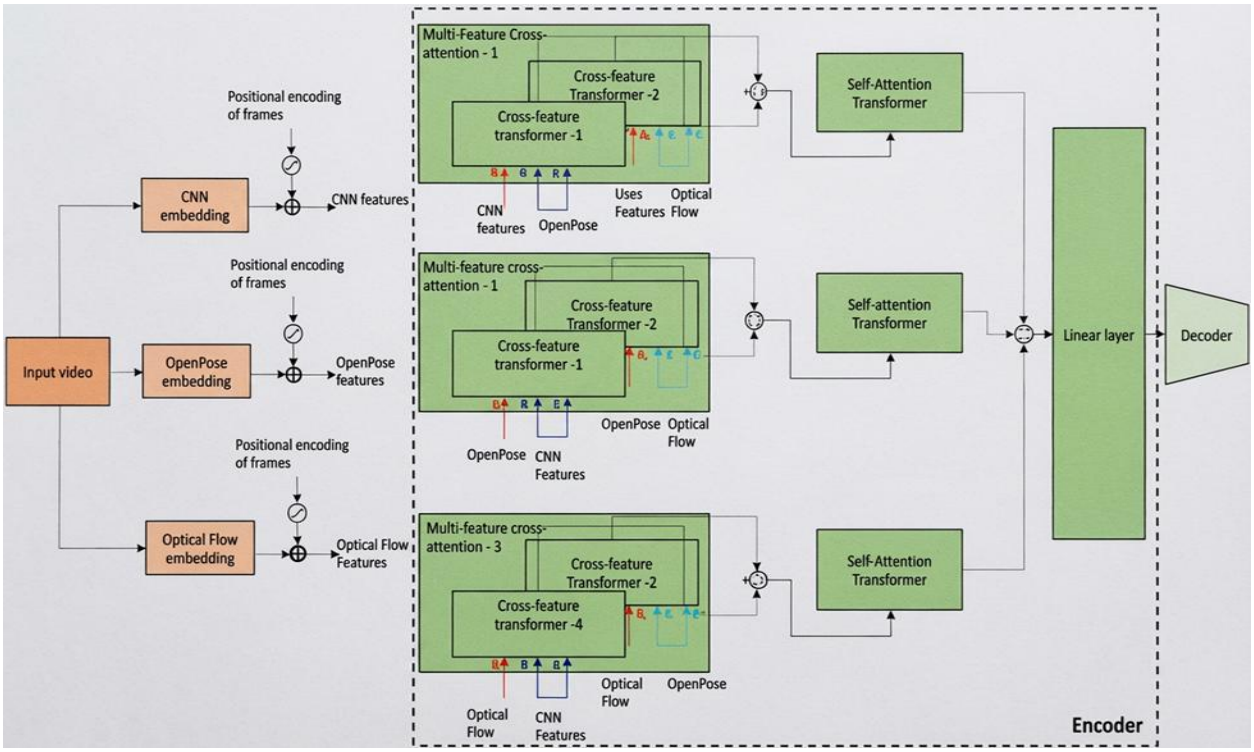
HARDWARE REQUIREMENTS:

- System : Pentium IV 2.4 GHz.
- Hard Disk : 40 GB.
- Ram : 512 Mb.

SOFTWARE REQUIREMENTS:

- Operating system : - Windows.
- Coding Language : python

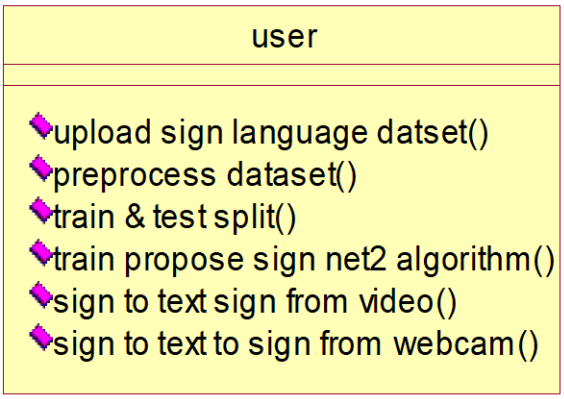
System Architecture



UML Diagrams:

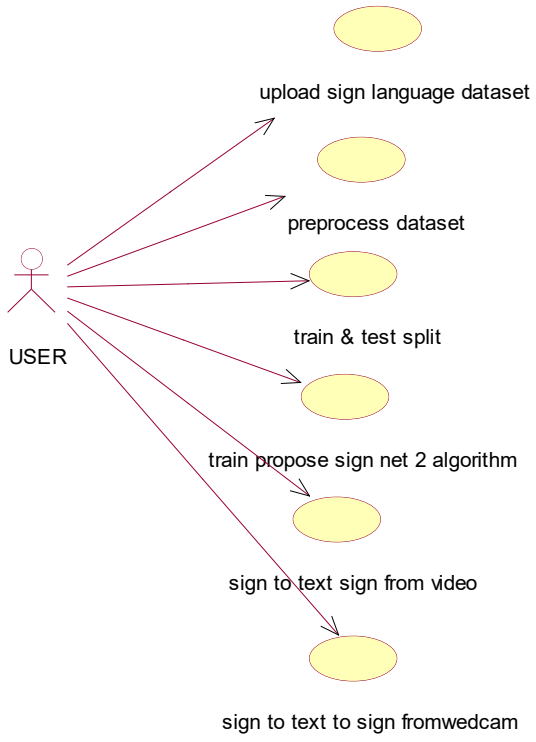
CLASS DIAGRAM:

The class diagram is used to refine the use case diagram and define a detailed design of the system. The class diagram classifies the actors defined in the use case diagram into a set of interrelated classes. The relationship or association between the classes can be either an "is-a" or "has-a" relationship. Each class in the class diagram may be capable of providing certain functionalities. These functionalities provided by the class are termed "methods" of the class. Apart from this, each class may have certain "attributes" that uniquely.



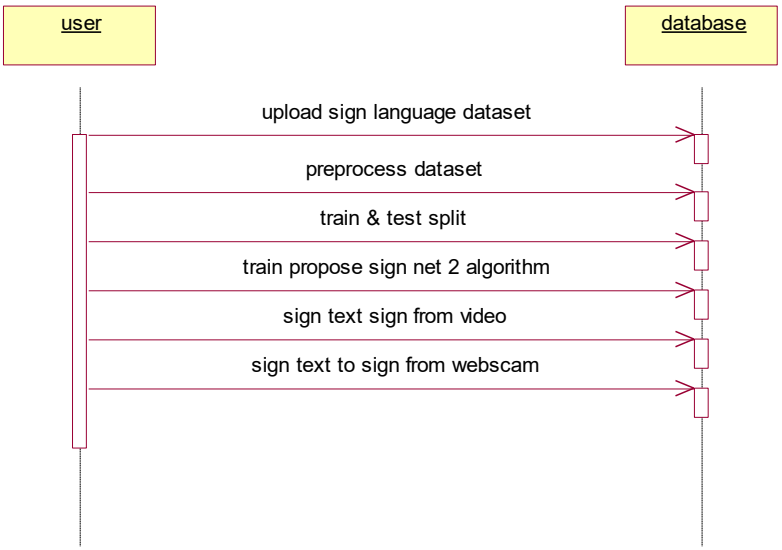
Use case Diagram:

A use case diagram in the Unified Modeling Language (UML) is a type of behavioral diagram defined by and created from a Use-case analysis. Its purpose is to present a graphical overview of the functionality provided by a system in terms of actors, their goals (represented as use cases), and any dependencies between those use cases. The main purpose of a use case diagram is to show what system functions are performed for which actor. Roles of the actors in the system can be depicted.



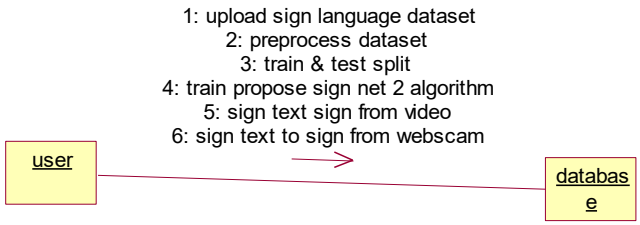
Sequence Diagram:

A sequence diagram represents the interaction between different objects in the system. The important aspect of a sequence diagram is that it is time-ordered. This means that the exact sequence of the interactions between the objects is represented step by step. Different objects in the sequence diagram interact with each other by passing "messages"



Collaborative Diagram:

A collaboration diagram groups together the interactions between different objects. The interactions are listed as numbered interactions that help to trace the sequence of the interactions. The collaboration diagram helps to identify all the possible interactions that each object has with other objects.



System Implementations:

The implementation of SIGNNET II involves several stages, from data preprocessing and model design to training, evaluation, and deployment. This section outlines each of these components and how they collectively contribute to the overall functionality of the two-way sign language translation system.

1. Data Collection and Preprocessing

- **Datasets Used:**

  Publicly available datasets such as RWTH-PHOENIX-Weather 2014T, CSL

(Chinese Sign Language), and ASLLVD (American Sign Language Lexicon Video Dataset) are utilized.

- **Preprocessing Steps:**
  - **Frame Extraction:** Sign language videos are decomposed into individual frames.
  - **Hand and Pose Detection:** OpenPose or MediaPipe is used to extract hand keypoints, body pose, and facial landmarks.
  - **Normalization:** Frame sizes and keypoint coordinates are normalized.
  - **Gloss Annotation Alignment:** Text labels or gloss annotations are aligned with frame sequences.

2. Model Architecture

- **Two-Way Transformer Model:**
  - **Encoder:**
    - For Sign-to-Text: A combination of CNN (for visual features) and self-attention layers captures spatial-temporal features.
    - For Text-to-Sign: Text tokens are embedded and passed through positional encoding layers.
  - **Decoder:**
    - For Sign-to-Text: Decodes encoded video features into coherent text using attention mechanisms.

        - For Text-to-Sign: Decodes text embeddings into sequences of gesture embeddings (used to generate glosses or avatar-driven signs).
    - **Attention Mechanism:** Multi-head attention layers are used to understand contextual relationships both spatially (in video) and linguistically (in text).

3. Training Procedure

- **Loss Functions:**
    - Cross-entropy loss for text generation.
    - Sequence alignment loss for gloss prediction.
    - Optional auxiliary losses for multimodal feature learning.
- **Optimization:**
    - Adam optimizer with a learning rate scheduler.
    - Gradient clipping to avoid exploding gradients.
- **Training Strategy:**
    - Batch processing with data augmentation (e.g., temporal jittering, flipping).
    - Use of validation data for early stopping and performance monitoring.

4. Testing and Evaluation

- **Metrics Used:**
    - BLEU and ROUGE scores for text translation.

    - Word Error Rate (WER) and Sign Error Rate (SER).
    - Real-time inference time for latency evaluation.
- **Testing Scenarios:**
    - Seen and unseen signers.
    - Varied lighting and background conditions.
    - Continuous and isolated signing evaluation.

5. Deployment

- **Model Compression:**
    - Quantization and pruning techniques are applied to reduce model size and inference time.
- **Interface:**
    - Web-based or mobile-friendly UI for uploading video/text and viewing results.
    - Optional integration with sign avatar systems for visual sign generation.
- **Real-Time Integration:**
    - Compatible with edge devices (e.g., Raspberry Pi, Jetson Nano).
    - Cloud-based APIs for scalable deployment in educational and public service applications.

System Environment:

What is Python :-

Below are some facts about Python.

Python is currently the most widely used multi-purpose, high-level programming language.

Python allows programming in Object-Oriented and Procedural paradigms. Python programs generally are smaller than other programming languages like Java.

Programmers have to type relatively less and indentation requirement of the language, makes them readable all the time.

Python language is being used by almost all tech-giant companies like – Google, Amazon, Facebook, Instagram, Dropbox, Uber… etc.

The biggest strength of Python is huge collection of standard library which can be used for the following .

- Machine Learning
- GUI Applications (like Kivy, Tkinter, PyQt etc.)
- Web frameworks like Django (used by YouTube, Instagram, Dropbox)
- Image processing (like Opencv, Pillow)
- Web scraping (like Scrapy, BeautifulSoup, Selenium)
- Test frameworks
- Multimedia

Advantages of Python :-

Let's see how Python dominates over other languages.

1. Extensive Libraries

Python downloads with an extensive library and it *contain code for various purposes like regular expressions, documentation-generation, unit-testing, web browsers, threading, databases, CGI, email, image manipulation, and more.* So, we don't have to write the complete code for that manually.

2. Extensible

As we have seen earlier, Python can be **extended to other languages**. You can write some of your code in languages like C++ or C. This comes in handy, especially in projects.

3. Embeddable

Complimentary to extensibility, Python is embeddable as well. You can put your Python code in your source code of a different language, like C++. This lets us add **scripting capabilities** to our code in the other language.

4. Improved Productivity

The language's simplicity and extensive libraries render programmers **more productive** than languages like Java and C++ do. Also, the fact that you need to write less and get more things done.

5. IOT Opportunities

Since Python forms the basis of new platforms like Raspberry Pi, it finds the future bright for the Internet Of Things. This is a way to connect the language with the real world.

6. Simple and Easy

When working with Java, you may have to create a class to print **'Hello World'**. But in Python, just a print statement will do. It is also quite **easy to learn, understand,** and **code.** This is why when people pick up Python, they have a hard time adjusting to other more verbose languages like Java.

7. Readable

Because it is not such a verbose language, reading Python is much like reading English. This is the reason why it is so easy to learn, understand, and code. It also does not need curly braces to define blocks, and **indentation is mandatory.** This further aids the readability of the code.

8. Object-Oriented

This language supports both the **procedural and object-oriented** programming paradigms. While functions help us with code reusability, classes and objects let us model the real world. A class allows the **encapsulation of data** and functions into one.

9. Free and Open-Source

Like we said earlier, Python is **freely available.** But not only can you **download Python** for free, but you can also download its source code, make changes to it, and even distribute it. It downloads with an extensive collection of libraries to help you with your tasks.

10. Portable

When you code your project in a language like C++, you may need to make some changes to it if you want to run it on another platform. But it isn't the same with Python. Here, you need to **code only once**, and you can run it anywhere. This is called **Write Once Run Anywhere (WORA)**. However, you need to be careful enough not to include any system-dependent features.

11. Interpreted

Lastly, we will say that it is an interpreted language. Since statements are executed one by one, **debugging is easier** than in compiled languages.
Any doubts till now in the advantages of Python? Mention in the comment section.

Advantages of Python Over Other Languages

1. Less Coding

Almost all of the tasks done in Python requires less coding when the same task is done in other languages. Python also has an awesome standard library support, so you don't have to search for any third-party libraries to get your job done. This is the reason that many people suggest learning Python to beginners.

2. Affordable

Python is free therefore individuals, small companies or big organizations can leverage the free available resources to build applications. Python is popular and widely used so it gives you better community support.

The 2019 Github annual survey showed us that Python has overtaken Java in the most popular programming language category.

3. Python is for Everyone

Python code can run on any machine whether it is Linux, Mac or Windows. Programmers need to learn different languages for different jobs but with Python, you can professionally build web apps, perform data analysis and **machine learning**, automate things, do web scraping and also build games and powerful visualizations. It is an all-rounder programming language.

Disadvantages of Python

So far, we've seen why Python is a great choice for your project. But if you choose it, you should be aware of its consequences as well. Let's now see the downsides of choosing Python over another language.

1. Speed Limitations

We have seen that Python code is executed line by line. But since Python is interpreted, it often results in **slow execution**. This, however, isn't a problem unless speed is a focal point for the project. In other words, unless high speed is

a requirement, the benefits offered by Python are enough to distract us from its speed limitations.

2. Weak in Mobile Computing and Browsers

While it serves as an excellent server-side language, Python is much rarely seen on the **client-side**. Besides that, it is rarely ever used to implement smartphone-based applications. One such application is called **Carbonnelle**.
The reason it is not so famous despite the existence of Brython is that it isn't that secure.

3. Design Restrictions

As you know, Python is **dynamically-typed**. This means that you don't need to declare the type of variable while writing the code. It uses **duck-typing**. But wait, what's that? Well, it just means that if it looks like a duck, it must be a duck. While this is easy on the programmers during coding, it can **raise run-time errors**.

4. Underdeveloped Database Access Layers

Compared to more widely used technologies like **JDBC (Java DataBase Connectivity)** and **ODBC (Open DataBase Connectivity)**, Python's database access layers are a bit underdeveloped. Consequently, it is less often applied in huge enterprises.

5. Simple

No, we're not kidding. Python's simplicity can indeed be a problem. Take my example. I don't do Java, I'm more of a Python person. To me, its syntax is so simple that the verbosity of Java code seems unnecessary.

This was all about the Advantages and Disadvantages of Python Programming Language.

History of Python : -

What do the alphabet and the programming language Python have in common? Right, both start with ABC. If we are talking about ABC in the Python context, it's clear that the programming language ABC is meant. ABC is a general-purpose programming language and programming environment, which had been developed in the Netherlands, Amsterdam, at the CWI (Centrum Wiskunde &Informatica). The greatest achievement of ABC was to influence the design of Python.Python was conceptualized in the late 1980s. Guido van Rossum worked that time in a project at the CWI, called Amoeba, a distributed operating system. In an interview with Bill Venners[1], Guido van Rossum said: "In the early 1980s, I worked as an implementer on a team building a language called ABC at Centrum voor Wiskunde en Informatica (CWI).

I don't know how well people know ABC's influence on Python. I try to mention ABC's influence because I'm indebted to everything I learned during that project and to the people who worked on it."Later on in the same Interview, Guido van Rossum continued: "I remembered all my experience and some of my frustration with ABC. I decided to try to design a simple scripting language that possessed some of ABC's better properties, but without its problems. So I started typing. I created a simple virtual machine, a simple parser, and a simple runtime. I made my own version of the various ABC parts that I liked. I created a basic syntax, used indentation for statement grouping instead of curly braces or begin-end

blocks, and developed a small number of powerful data types: a hash table (or dictionary, as we call it), a list, strings, and numbers."

What is Machine Learning : -

Before we take a look at the details of various machine learning methods, let's start by looking at what machine learning is, and what it isn't. Machine learning is often categorized as a subfield of artificial intelligence, but I find that categorization can often be misleading at first brush. The study of machine learning certainly arose from research in this context, but in the data science application of machine learning methods, it's more helpful to think of machine learning as a means of *building models of data*.

Fundamentally, machine learning involves building mathematical models to help understand data. "Learning" enters the fray when we give these models *tunable parameters* that can be adapted to observed data; in this way the program can be considered to be "learning" from the data.

Once these models have been fit to previously seen data, they can be used to predict and understand aspects of newly observed data. I'll leave to the reader the more philosophical digression regarding the extent to which this type of mathematical, model-based "learning" is similar to the "learning" exhibited by the human brain.Understanding the problem setting in machine learning is essential to using these tools effectively, and so we will start with some broad categorizations of the types of approaches we'll discuss here.

Categories Of Machine Leaning :-

At the most fundamental level, machine learning can be categorized into two main types: supervised learning and unsupervised learning.

Supervised learning involves somehow modeling the relationship between measured features of data and some label associated with the data; once this model is determined, it can be used to apply labels to new, unknown data. This is further subdivided into *classification* tasks and *regression* tasks: in classification, the labels are discrete categories, while in regression, the labels are continuous quantities. We will see examples of both types of supervised learning in the following section.

Unsupervised learning involves modeling the features of a dataset without reference to any label, and is often described as "letting the dataset speak for itself." These models include tasks such as *clustering* and *dimensionality reduction.*

Clustering algorithms identify distinct groups of data, while dimensionality reduction algorithms search for more succinct representations of the data. We will see examples of both types of unsupervised learning in the following section.

Need for Machine Learning

Human beings, at this moment, are the most intelligent and advanced species on earth because they can think, evaluate and solve complex problems. On the other side, AI is still in its initial stage and haven’t surpassed human intelligence in many aspects. Then the question is that what is the need to make machine learn?

The most suitable reason for doing this is, “to make decisions, based on data, with efficiency and scale”.

Lately, organizations are investing heavily in newer technologies like Artificial Intelligence, Machine Learning and Deep Learning to get the key information from data to perform several real-world tasks and solve problems. We can call it data-driven decisions taken by machines, particularly to automate the process. These data-driven decisions can be used, instead of using programing logic, in the problems that cannot be programmed inherently. The fact is that we can’t do without human intelligence, but other aspect is that we all need to solve real-world problems with efficiency at a huge scale. That is why the need for machine learning arises.

Challenges in Machines Learning :-

While Machine Learning is rapidly evolving, making significant strides with cybersecurity and autonomous cars, this segment of AI as whole still has a long way to go. The reason behind is that ML has not been able to overcome number of challenges. The challenges that ML is facing currently are −

Quality of data − Having good-quality data for ML algorithms is one of the biggest challenges. Use of low-quality data leads to the problems related to data preprocessing and feature extraction.

Time-Consuming task − Another challenge faced by ML models is the consumption of time especially for data acquisition, feature extraction and retrieval.

Lack of specialist persons − As ML technology is still in its infancy stage, availability of expert resources is a tough job.

No clear objective for formulating business problems − Having no clear objective and well-defined goal for business problems is another key challenge for ML because this technology is not that mature yet.

Issue of overfitting & underfitting − If the model is overfitting or underfitting, it cannot be represented well for the problem.

Curse of dimensionality − Another challenge ML model faces is too many features of data points. This can be a real hindrance.

Difficulty in deployment − Complexity of the ML model makes it quite difficult to be deployed in real life.

Applications of Machines Learning :-

Machine Learning is the most rapidly growing technology and according to researchers we are in the golden year of AI and ML. It is used to solve many real-world complex problems which cannot be solved with traditional approach. Following are some real-world applications of ML −

- Emotion analysis
- Sentiment analysis
- Error detection and prevention
- Weather forecasting and prediction
- Stock market analysis and forecasting
- Speech synthesis

- Speech recognition
- Customer segmentation
- Object recognition
- Fraud detection
- Fraud prevention
- Recommendation of products to customer in online shopping

How to Start Learning Machine Learning?

Arthur Samuel coined the term **"Machine Learning"** in 1959 and defined it as a **"Field of study that gives computers the capability to learn without being explicitly programmed".**

And that was the beginning of Machine Learning! In modern times, Machine Learning is one of the most popular (if not the most!) career choices. According to Indeed, Machine Learning Engineer Is The Best Job of 2019 with a *344%* growth and an average base salary of **$146,085** per year.

But there is still a lot of doubt about what exactly is Machine Learning and how to start learning it? So this article deals with the Basics of Machine Learning and also the path you can follow to eventually become a full-fledged Machine Learning Engineer. Now let's get started!!!

How to start learning ML?

This is a rough roadmap you can follow on your way to becoming an insanely talented Machine Learning Engineer. Of course, you can always modify the steps according to your needs to reach your desired end-goal!

Step 1 – Understand the Prerequisites

In case you are a genius, you could start ML directly but normally, there are some prerequisites that you need to know which include Linear Algebra, Multivariate Calculus, Statistics, and Python. And if you don't know these, never fear! You don't need a Ph.D. degree in these topics to get started but you do need a basic understanding.

(a) Learn Linear Algebra and Multivariate Calculus

Both Linear Algebra and Multivariate Calculus are important in Machine Learning. However, the extent to which you need them depends on your role as a data scientist. If you are more focused on application heavy machine learning, then you will not be that heavily focused on maths as there are many common libraries available. But if you want to focus on R&D in Machine Learning, then mastery of Linear Algebra and Multivariate Calculus is very important as you will have to implement many ML algorithms from scratch.

(b) Learn Statistics

Data plays a huge role in Machine Learning. In fact, around 80% of your time as an ML expert will be spent collecting and cleaning data. And statistics is a field

that handles the collection, analysis, and presentation of data. So it is no surprise that you need to learn it!!!

Some of the key concepts in statistics that are important are Statistical Significance, Probability Distributions, Hypothesis Testing, Regression, etc. Also, Bayesian Thinking is also a very important part of ML which deals with various concepts like Conditional Probability, Priors, and Posteriors, Maximum Likelihood, etc.

(c) Learn Python

Some people prefer to skip Linear Algebra, Multivariate Calculus and Statistics and learn them as they go along with trial and error. But the one thing that you absolutely cannot skip is Python! While there are other languages you can use for Machine Learning like R, Scala, etc. Python is currently the most popular language for ML. In fact, there are many Python libraries that are specifically useful for Artificial Intelligence and Machine Learning such as Keras, TensorFlow, Scikit-learn, etc.

So if you want to learn ML, it's best if you learn Python! You can do that using various online resources and courses such as **Fork Python** available Free on GeeksforGeeks.

Step 2 – Learn Various ML Concepts

Now that you are done with the prerequisites, you can move on to actually learning ML (Which is the fun part!!!) It's best to start with the basics and then move on to the more complicated stuff. Some of the basic concepts in ML are:

(a) Terminologies of Machine Learning

- **Model –** A model is a specific representation learned from data by applying some machine learning algorithm. A model is also called a hypothesis.
- **Feature –** A feature is an individual measurable property of the data. A set of numeric features can be conveniently described by a feature vector. Feature vectors are fed as input to the model. For example, in order to predict a fruit, there may be features like color, smell, taste, etc.
- **Target (Label) –** A target variable or label is the value to be predicted by our model. For the fruit example discussed in the feature section, the label with each set of input would be the name of the fruit like apple, orange, banana, etc.
- **Training –** The idea is to give a set of inputs(features) and it's expected outputs(labels), so after training, we will have a model (hypothesis) that will then map new data to one of the categories trained on.
- **Prediction –** Once our model is ready, it can be fed a set of inputs to which it will provide a predicted output(label).

(b) Types of Machine Learning

- **Supervised Learning –** This involves learning from a training dataset with labeled data using classification and regression models. This learning process continues until the required level of performance is achieved.

- **Unsupervised Learning –** This involves using unlabelled data and then finding the underlying structure in the data in order to learn more and more about the data itself using factor and cluster analysis models.
- **Semi-supervised Learning –** This involves using unlabelled data like Unsupervised Learning with a small amount of labeled data. Using labeled data vastly increases the learning accuracy and is also more cost-effective than Supervised Learning.
- **Reinforcement Learning –** This involves learning optimal actions through trial and error. So the next action is decided by learning behaviors that are based on the current state and that will maximize the reward in the future.

Advantages of Machine learning :-

1. Easily identifies trends and patterns -

Machine Learning can review large volumes of data and discover specific trends and patterns that would not be apparent to humans. For instance, for an e-commerce website like Amazon, it serves to understand the browsing behaviors and purchase histories of its users to help cater to the right products, deals, and reminders relevant to them. It uses the results to reveal relevant advertisements to them.

2. No human intervention needed (automation)

With ML, you don't need to babysit your project every step of the way. Since it means giving machines the ability to learn, it lets them make predictions and also improve the algorithms on their own. A common example of this is anti-virus softwares; they learn to filter new threats as they are recognized. ML is also good at recognizing spam.

3. Continuous Improvement

As **ML algorithms** gain experience, they keep improving in accuracy and efficiency. This lets them make better decisions. Say you need to make a weather forecast model. As the amount of data you have keeps growing, your algorithms learn to make more accurate predictions faster.

4. Handling multi-dimensional and multi-variety data

Machine Learning algorithms are good at handling data that are multi-dimensional and multi-variety, and they can do this in dynamic or uncertain environments.

5. Wide Applications

You could be an e-tailer or a healthcare provider and make ML work for you. Where it does apply, it holds the capability to help deliver a much more personal experience to customers while also targeting the right customers.

Disadvantages of Machine Learning :-

1. Data Acquisition

Machine Learning requires massive data sets to train on, and these should be inclusive/unbiased, and of good quality. There can also be times where they must wait for new data to be generated.

2. Time and Resources

ML needs enough time to let the algorithms learn and develop enough to fulfill their purpose with a considerable amount of accuracy and relevancy. It also needs massive resources to function. This can mean additional requirements of computer power for you.

3. Interpretation of Results

Another major challenge is the ability to accurately interpret results generated by the algorithms. You must also carefully choose the algorithms for your purpose.

4. High error-susceptibility

Machine Learning is autonomous but highly susceptible to errors. Suppose you train an algorithm with data sets small enough to not be inclusive. You end up with biased predictions coming from a biased training set. This leads to irrelevant advertisements being displayed to customers. In the case of ML, such blunders can set off a chain of errors that can go undetected for long periods of time. And when they do get noticed, it takes quite some time to recognize the source of the issue, and even longer to correct it.

Python Development Steps : -

Guido Van Rossum published the first version of Python code (version 0.9.0) at alt.sources in February 1991. This release included already exception handling, functions, and the core data types of list, dict, str and others. It was also object oriented and had a module system.

Python version 1.0 was released in January 1994. The major new features included in this release were the functional programming tools lambda, map, filter and reduce, which Guido Van Rossum never liked.Six and a half years later in October 2000, Python 2.0 was introduced. This release included list comprehensions, a full garbage collector and it was supporting unicode.Python flourished for another 8 years in the versions 2.x before the next major release as Python 3.0 (also known

as "Python 3000" and "Py3K") was released. Python 3 is not backwards compatible with Python 2.x.

The emphasis in Python 3 had been on the removal of duplicate programming constructs and modules, thus fulfilling or coming close to fulfilling the 13th law of the Zen of Python: "There should be one -- and preferably only one -- obvious way to do it."Some changes in Python 7.3:

- Print is now a function
- Views and iterators instead of lists
- The rules for ordering comparisons have been simplified. E.g. a heterogeneous list cannot be sorted, because all the elements of a list must be comparable to each other.
- There is only one integer type left, i.e. int. long is int as well.
- The division of two integers returns a float instead of an integer. "//" can be used to have the "old" behaviour.
- Text Vs. Data Instead Of Unicode Vs. 8-bit

Purpose :-

We demonstrated that our approach enables successful segmentation of intra-retinal layers—even with low-quality images containing speckle noise, low contrast, and different intensity ranges throughout—with the assistance of the ANIS feature.

Python

Python is an interpreted high-level programming language for general-purpose programming. Created by Guido van Rossum and first released in 1991, Python has a design philosophy that emphasizes code readability, notably using significant whitespace.

Python features a dynamic type system and automatic memory management. It supports multiple programming paradigms, including object-oriented, imperative, functional and procedural, and has a large and comprehensive standard library.

- Python is Interpreted − Python is processed at runtime by the interpreter. You do not need to compile your program before executing it. This is similar to PERL and PHP.
- Python is Interactive − you can actually sit at a Python prompt and interact with the interpreter directly to write your programs.

Python also acknowledges that speed of development is important. Readable and terse code is part of this, and so is access to powerful constructs that avoid tedious repetition of code. Maintainability also ties into this may be an all but useless metric, but it does say something about how much code you have to scan, read and/or understand to troubleshoot problems or tweak behaviors. This speed of development, the ease with which a programmer of other languages can pick up basic Python skills and the huge standard library is key to another area where Python excels. All its tools have been quick to implement, saved a lot of time,

and several of them have later been patched and updated by people with no Python background - without breaking.

Modules Used in Project :-

Tensorflow

TensorFlow is a free and open-source software library for dataflow and differentiable programming across a range of tasks. It is a symbolic math library, and is also used for machine learning applications such as neural networks. It is used for both research and production at Google.

TensorFlow was developed by the Google Brain team for internal Google use. It was released under the Apache 2.0 open-source license on November 9, 2015.

Numpy

Numpy is a general-purpose array-processing package. It provides a high-performance multidimensional array object, and tools for working with these arrays.

It is the fundamental package for scientific computing with Python. It contains various features including these important ones:

- A powerful N-dimensional array object
- Sophisticated (broadcasting) functions
- Tools for integrating C/C++ and Fortran code
- Useful linear algebra, Fourier transform, and random number capabilities

Besides its obvious scientific uses, Numpy can also be used as an efficient multi-dimensional container of generic data. Arbitrary data-types can be defined using

Numpy which allows Numpy to seamlessly and speedily integrate with a wide variety of databases.

Pandas

Pandas is an open-source Python Library providing high-performance data manipulation and analysis tool using its powerful data structures. Python was majorly used for data munging and preparation. It had very little contribution towards data analysis. Pandas solved this problem. Using Pandas, we can accomplish five typical steps in the processing and analysis of data, regardless of the origin of data load, prepare, manipulate, model, and analyze. Python with Pandas is used in a wide range of fields including academic and commercial domains including finance, economics, Statistics, analytics, etc.

Matplotlib

Matplotlib is a Python 2D plotting library which produces publication quality figures in a variety of hardcopy formats and interactive environments across platforms. Matplotlib can be used in Python scripts, the Python and IPython shells, the Jupyter Notebook, web application servers, and four graphical user interface toolkits. Matplotlib tries to make easy things easy and hard things possible. You can generate plots, histograms, power spectra, bar charts, error charts, scatter plots, etc., with just a few lines of code. For examples, see the sample plots and thumbnail gallery.

For simple plotting the pyplot module provides a MATLAB-like interface, particularly when combined with IPython. For the power user, you have full control of line styles, font properties, axes properties, etc, via an object oriented interface or via a set of functions familiar to MATLAB users.

Scikit – learn

Scikit-learn provides a range of supervised and unsupervised learning algorithms via a consistent interface in Python. It is licensed under a permissive simplified BSD license and is distributed under many Linux distributions, encouraging academic and commercial use. **Python**

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  All its tools have been quick to implement, saved a lot of time, and several of them have later been patched and updated by people with no Python background - without breaking.

Install Python Step-by-Step in Windows and Mac :

Python a versatile programming language doesn't come pre-installed on your computer devices. Python was first released in the year 1991 and until today it is a very popular high-level programming language. Its style philosophy emphasizes code readability with its notable use of great whitespace.

The object-oriented approach and language construct provided by Python enables programmers to write both clear and logical code for projects. This software does not come pre-packaged with Windows.

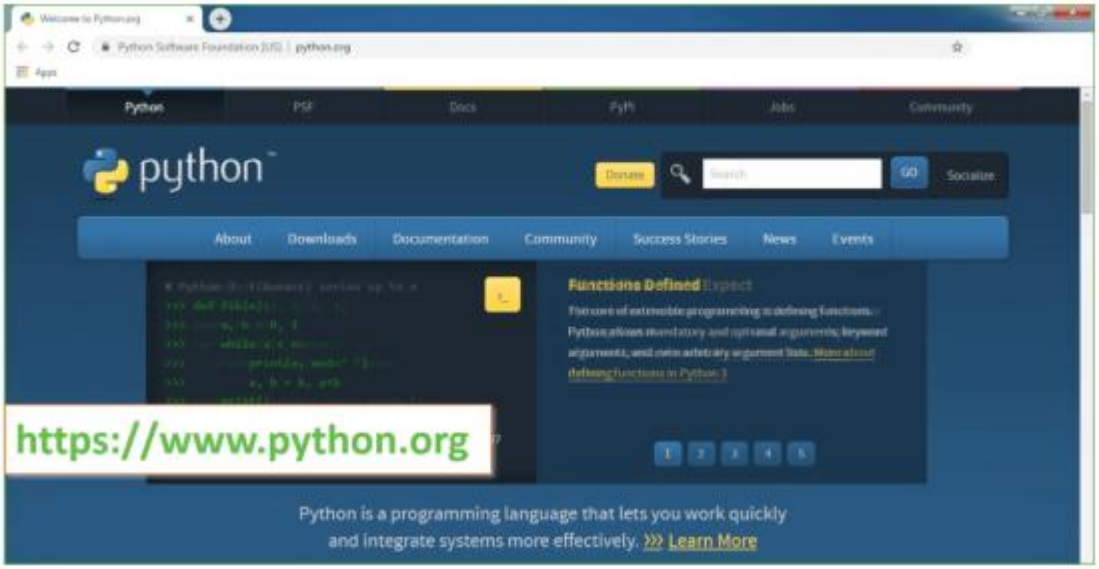
How to Install Python on Windows and Mac :

There have been several updates in the Python version over the years. The question is how to install Python? It might be confusing for the beginner who is willing to start learning Python but this tutorial will solve your query. The latest or the newest version of Python is version 3.7.4 or in other words, it is Python 3.
Note: The python version 3.7.4 cannot be used on Windows XP or earlier devices.

Before you start with the installation process of Python. First, you need to know about your **System Requirements**. Based on your system type i.e. operating system and based processor, you must download the python version. My system type is a **Windows 64-bit operating system**. So the steps below are to install python version 3.7.4 on Windows 7 device or to install Python 3. Download the Python Cheatsheet here.The steps on how to install Python on Windows 10, 8 and 7 are **divided into 4 parts** to help understand better.

Download the Correct version into the system

Step 1: Go to the official site to download and install python using Google Chrome or any other web browser. OR Click on the following link: https://www.python.org



Now, check for the latest and the correct version for your operating system.

Step 2: Click on the Download Tab.

Step 3: You can either select the Download Python for windows 3.7.4 button in Yellow Color or you can scroll further down and click on download with respective

to their version. Here, we are downloading the most recent python version for windows 3.7.4

Looking for a specific release?

Python releases by version number:

Release version	Release date		Click for more
Python 3.7.4	July 8, 2019	Download	Release Notes
Python 3.6.9	July 2, 2019	Download	Release Notes
Python 3.7.3	March 25, 2019	Download	Release Notes
Python 3.4.10	March 18, 2019	Download	Release Notes
Python 3.5.7	March 18, 2019	Download	Release Notes
Python 2.7.16	March 4, 2019	Download	Release Notes
Python 3.7.2	Dec. 24, 2018	Download	Release Notes

Step 4: Scroll down the page until you find the Files option.

Step 5: Here you see a different version of python along with the operating system.

Files

Version	Operating System	Description	MD5 Sum	File Size	GPG
Gzipped source tarball	Source release		68111671e5b2db4aef7b9ab018f0f9be	23017663	SIG
XZ compressed source tarball	Source release		d33e4aae66097051c2eca45ee3604803	17131432	SIG
macOS 64-bit/32-bit installer	Mac OS X	for Mac OS X 10.6 and later	6428b4fa7583daff1a442cbabcee08e6	34898416	SIG
macOS 64-bit installer	Mac OS X	for OS X 10.9 and later	5dd605c38217a45773bf5e4a936b241f	28082845	SIG
Windows help file	Windows		d639995734a2c06b2ac56cade6b4f7cd2	8131761	SIG
Windows x86-64 embeddable zip file	Windows	for AMD64/EM64T/x64	9b00c3cf6d9ec0b6abe83184a4072fa2	7504391	SIG
Windows x86-64 executable installer	Windows	for AMD64/EM64T/x64	a702b4b0ad76debdb3043a583e563400	26680368	SIG
Windows x86-64 web-based installer	Windows	for AMD64/EM64T/x64	28cb1c608bbd73ae8e53a3bd351b4bd2	1362904	SIG
Windows x86 embeddable zip file	Windows		9fab3b81f8b41879fda94133574139d8	6741628	SIG
Windows x86 executable installer	Windows		33cc802942a54446a3d8451476394789	25663848	SIG
Windows x86 web-based installer	Windows		1b670cfa5d317df82c30983ea371d87c	1324608	SIG

• To download Windows 32-bit python, you can select any one from the three options: Windows x86 embeddable zip file, Windows x86 executable installer or Windows x86 web-based installer.

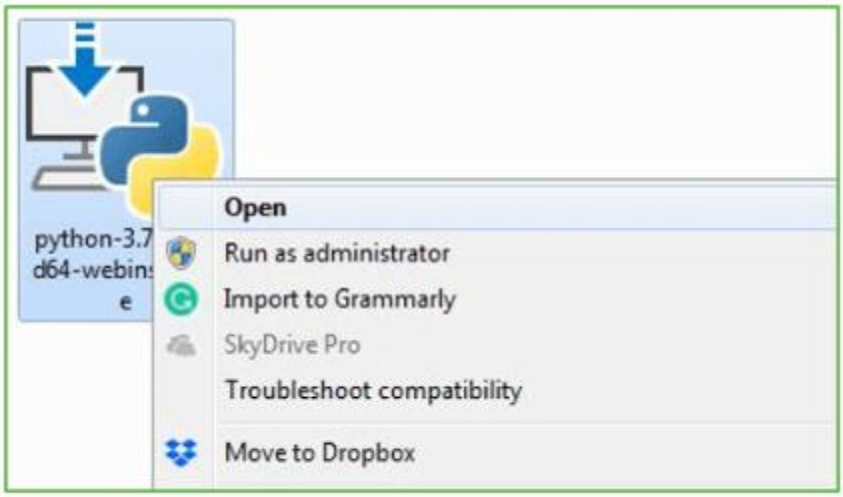
•To download Windows 64-bit python, you can select any one from the three options: Windows x86-64 embeddable zip file, Windows x86-64 executable installer or Windows x86-64 web-based installer.

Here we will install Windows x86-64 web-based installer. Here your first part regarding which version of python is to be downloaded is completed. Now we move ahead with the second part in installing python i.e. Installation

Note: To know the changes or updates that are made in the version you can click on the Release Note Option.

Installation of Python

Step 1: Go to Download and Open the downloaded python version to carry out the installation process.



Step 2: Before you click on Install Now, Make sure to put a tick on Add Python 3.7 to PATH.

Step 3: Click on Install NOW After the installation is successful. Click on Close.

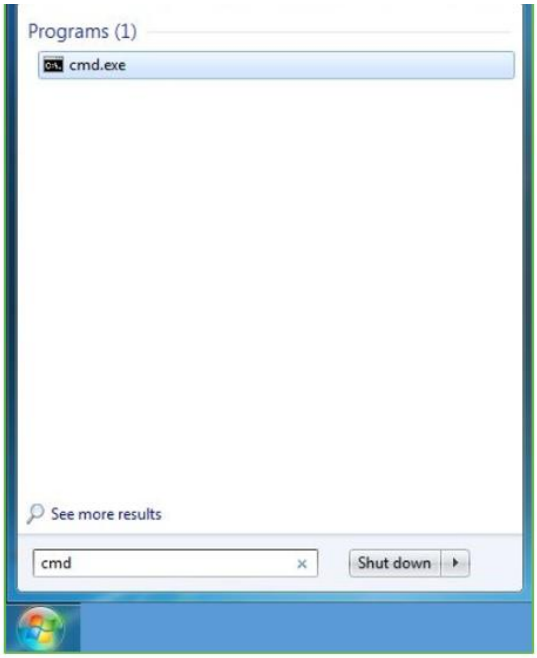
With these above three steps on python installation, you have successfully and correctly installed Python. Now is the time to verify the installation.

Note: The installation process might take a couple of minutes.

Verify the Python Installation

Step 1: Click on Start

Step 2: In the Windows Run Command, type “cmd”.



Step 3: Open the Command prompt option.

Step 4: Let us test whether the python is correctly installed. Type **python –V** and press Enter.

```
C:\Users\adity>python
Python 3.10.0 (tags/v3.10.0:b494f59, Oct  4 2021, 19:00:18) [MSC v.1929 64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license" for more information.
```

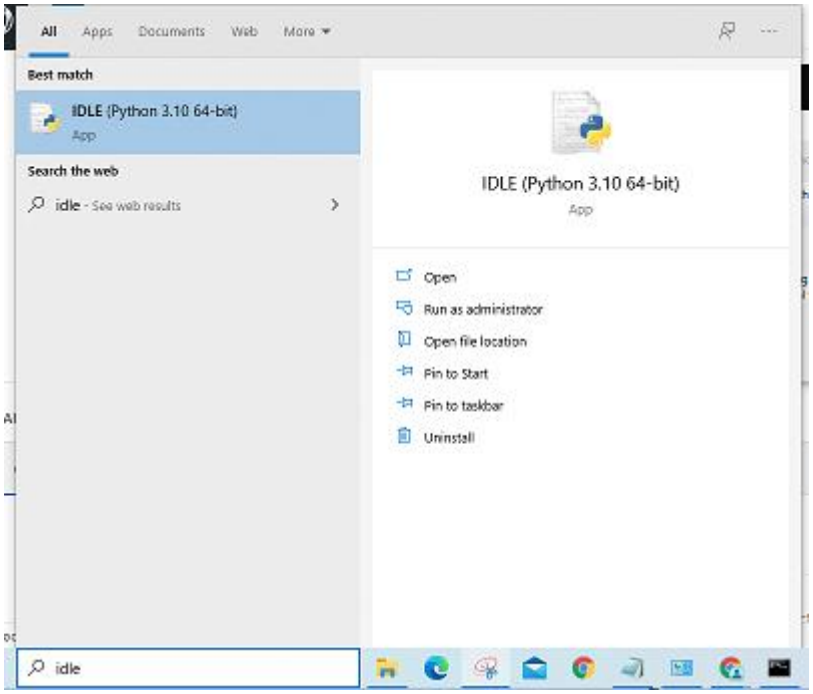
Step 5: You will get the answer as 3.10.0

Note: If you have any of the earlier versions of Python already installed. You must first uninstall the earlier version and then install the new one.

Check how the Python IDLE works

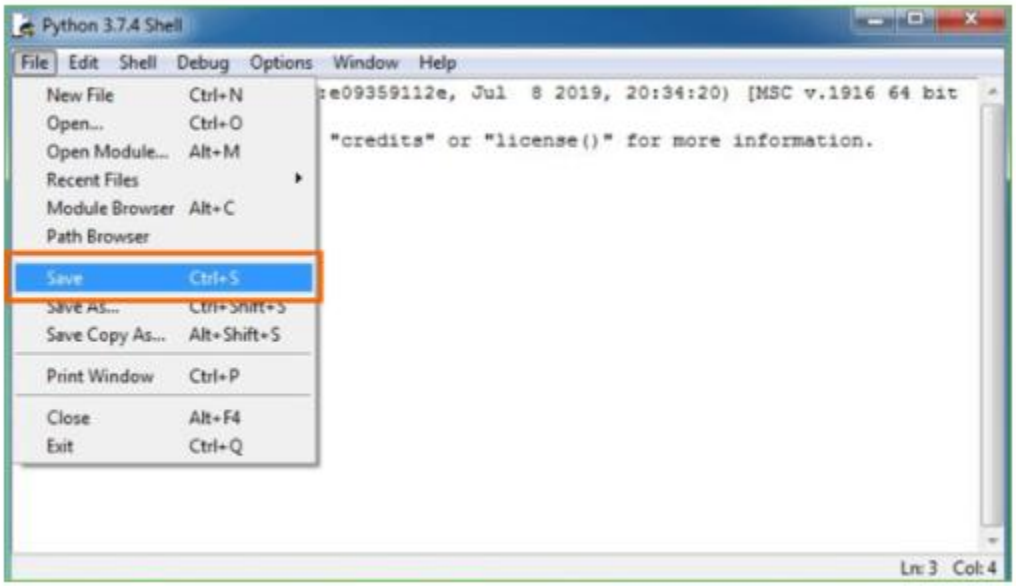
Step 1: Click on Start

Step 2: In the Windows Run command, type “python idle”.



Step 3: Click on IDLE (Python 3.10.0 64-bit) and launch the program

Step 4: To go ahead with working in IDLE you must first save the file. **Click on File > Click on Save**



Step 5: Name the file and save as type should be Python files. Click on SAVE. Here I have named the files as Hey World.

Step 6: Now for e.g. **enter print**

SYSTEM TEST

The purpose of testing is to discover errors. Testing is the process of trying to discover every conceivable fault or weakness in a work product. It provides a way to check the functionality of components, sub assemblies, assemblies and/or a finished product It is the process of exercising software with the intent of ensuring that the Software system meets its requirements and user expectations and does not fail in an unacceptable manner. There are various types of test. Each test type addresses a specific testing requirement.

TYPES OF TESTS

Unit testing

Unit testing involves the design of test cases that validate that the internal program logic is functioning properly, and that program inputs produce valid outputs. All decision branches and internal code flow should be validated. It is the testing of individual software units of the application .it is done after the completion of an individual unit before integration. This is a structural testing, that relies on knowledge of its construction and is invasive. Unit tests perform basic tests at component level and test a specific business process, application, and/or system configuration. Unit tests ensure that each unique path of a business process performs accurately to the documented specifications and contains clearly defined inputs and expected results.

Integration testing

Integration tests are designed to test integrated software components to determine if they actually run as one program. Testing is event driven and is more concerned with the basic outcome of screens or fields. Integration tests demonstrate that although the components were individually satisfaction, as shown by successfully unit testing, the combination of components is correct and consistent. Integration testing is specifically aimed at exposing the problems that arise from the combination of components.

Functional test

Functional tests provide systematic demonstrations that functions tested are available as specified by the business and technical requirements, system documentation, and user manuals.

Functional testing is centered on the following items:

Valid Input : identified classes of valid input must be accepted.

Invalid Input : identified classes of invalid input must be rejected.

Functions : identified functions must be exercised.

Output : identified classes of application outputs must be exercised.

Systems/Procedures : interfacing systems or procedures must be invoked.

Organization and preparation of functional tests is focused on requirements, key functions, or special test cases. In addition, systematic coverage pertaining to identify Business process flows; data fields, predefined processes, and successive processes must be considered for testing. Before functional testing is complete, additional tests are identified and the effective value of current tests is determined.

System Test

System testing ensures that the entire integrated software system meets requirements. It tests a configuration to ensure known and predictable results. An example of system testing is the configuration oriented system integration test. System testing is based on process descriptions and flows, emphasizing pre-driven process links and integration points.

White Box Testing

White Box Testing is a testing in which in which the software tester has knowledge of the inner workings, structure and language of the software, or at least its purpose. It is purpose. It is used to test areas that cannot be reached from a black box level.

Black Box Testing

Black Box Testing is testing the software without any knowledge of the inner workings, structure or language of the module being tested. Black box tests, as most other kinds of tests, must be written from a definitive source document, such as specification or requirements document, such as specification or requirements document. It is a testing in which the software under test is treated, as a black box .you cannot “see” into it. The test provides inputs and responds to outputs without considering how the software works.

Unit Testing

Unit testing is usually conducted as part of a combined code and unit test phase of the software lifecycle, although it is not uncommon for coding and unit testing to be conducted as two distinct phases.

Test strategy and approach

Field testing will be performed manually and functional tests will be written in detail.

Test objectives

- All field entries must work properly.
- Pages must be activated from the identified link.
- The entry screen, messages and responses must not be delayed.

Features to be tested

- Verify that the entries are of the correct format
- No duplicate entries should be allowed
- All links should take the user to the correct page.

Integration Testing

Software integration testing is the incremental integration testing of two or more integrated software components on a single platform to produce failures caused by interface defects.

The task of the integration test is to check that components or software applications, e.g. components in a software system or – one step up – software applications at the company level – interact without error.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Acceptance Testing

User Acceptance Testing is a critical phase of any project and requires significant participation by the end user. It also ensures that the system meets the functional requirements.

Test Results: All the test cases mentioned above passed successfully. No defects encountered.

Test cases1:

Test case for Login form:

FUNCTION:	**LOGIN**
EXPECTED RESULTS:	Should Validate the user and check his existence in database
ACTUAL RESULTS:	Validate the user and checking the user against the database
LOW PRIORITY	**No**
HIGH PRIORITY	**Yes**

Test case2:

Test case for User Registration form:

FUNCTION:	**USER REGISTRATION**
EXPECTED RESULTS:	Should check if all the fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are field by user or not through validations and saving user.
LOW PRIORITY	**No**
HIGH PRIORITY	**Yes**

Test case3:

Test case for Change Password:

When the old password does not match with the new password ,then this results in displaying an error message as " OLD PASSWORD DOES NOT MATCH WITH THE NEW PASSWORD".

FUNCTION:	**Change Password**
EXPECTED RESULTS:	Should check if old password and new password fields are filled by the user and saving the user to database.
ACTUAL RESULTS:	Checking whether all the fields are field by user or not through validations and saving user.
LOW PRIORITY	**No**
HIGH PRIORITY	**Yes**

SCREEN SHOTS

SignNet II: A Transformer-Based Two-Way Sign Language Translation Model

In propose paper author introducing SignNet fusion model which consists of two different Transformer Attention CNN Based encoder decoder models where first encoder-decoder model can get trained on Sign Images and Words and second encoder-model will take predicted TEXT output from first model to predict Sign image. So propose model will predict sign text from input video and then convert that text into sign image. In simple terms it's called as "Sing 2 Text and Text to Sign".

In propose paper author has used German and American language dataset but you ask us to use word level dataset so we have Word Based Sign dataset which consists of 18 different sign words showing below.

['I', 'apple', 'can', 'get', 'good', 'have', 'help', 'how', 'like', 'love', 'my', 'no', 'sorry', 'thank-you', 'want', 'yes', 'you', 'your']

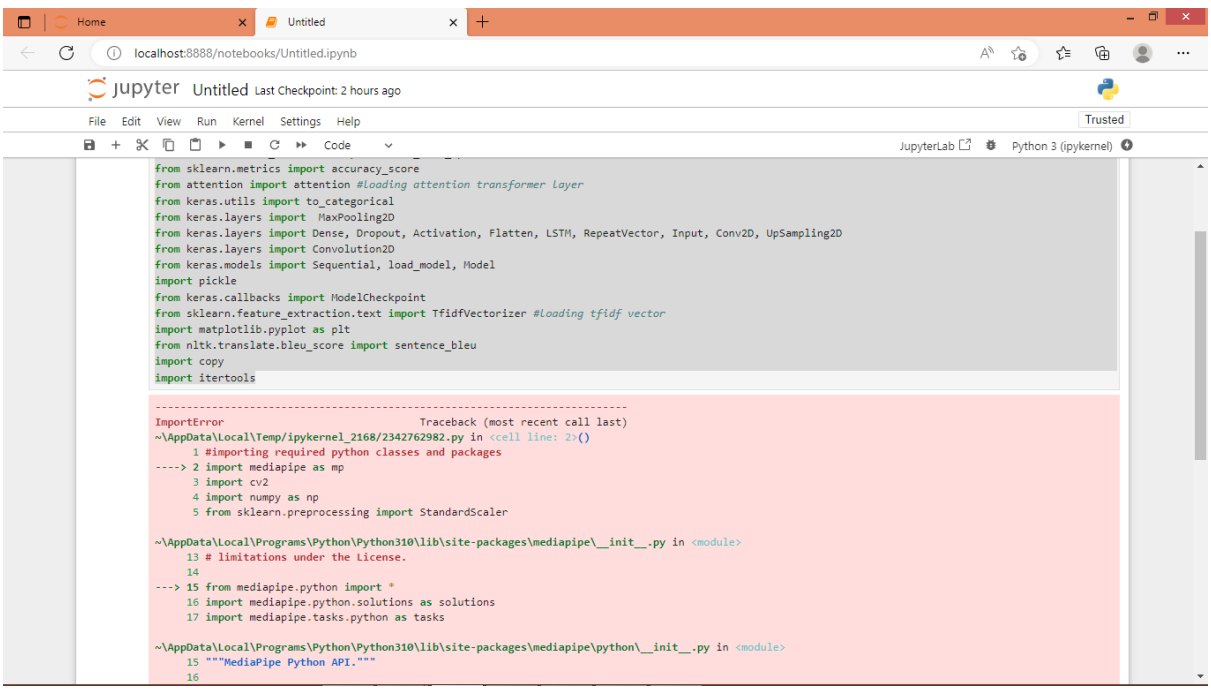
Above dataset can be downloading from below link

https://universe.roboflow.com/asl-dataset/asl-dataset-p9yw8

To extract features from video frames author has used Open Pose algorithm to detect hand and finger key-points and then utilize word2vec model to convert words into numeric vector. All extracted features will be input to two-way encoder decoder algorithm to train a model and this model can be applied on Live Webcam or video framers to predict sign text to sign image,

Author evaluated propose model by using BLEU score which takes predicted and true words as input and then calculate percentage of number of correct words prediction.

Note: we tried to train above model using JUPYTER notebook but JUPYTER API is unable to load open pose model to extract hand and finger key points so we have implemented this project using TKINTER GUI mode. In below screen showing error giving by JUPYTER



In above screen you can see JUPYTER is unable to load Media-pipe open pose model for hands and finger features extraction.

To implement this project we have designed following modules

1) Upload Sign Language Dataset: using this module user can upload dataset and then application will read all images and text and then extract hand and finger key points and convert text to numeric vector

2) Pre-process Dataset: all extracted features will be shuffle and normalize

3) Train & Test Split: split process data into train and test where application using 80% data for training and 20% for testing

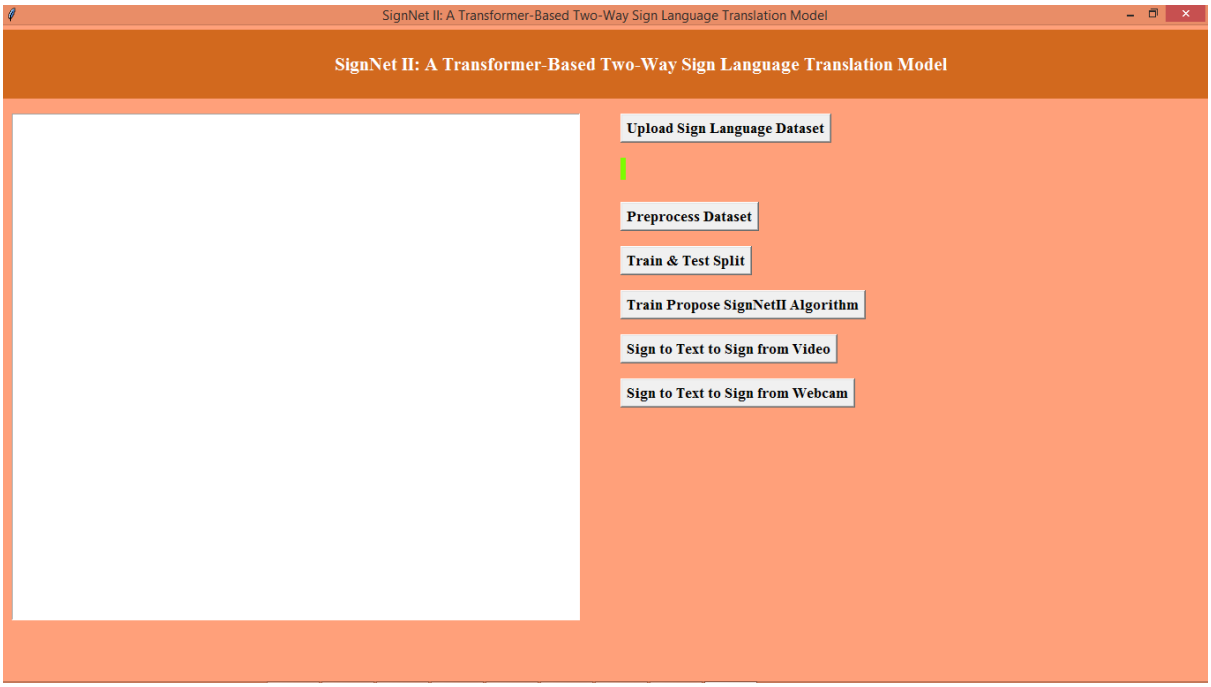
4) Train Propose SignNetII Algorithm: 80% training data will be input to SignNet algorithm to train a model and this model will be applied on 20% test data to calculate BLEU score which will be higher than propose model

5) Sign to Text to Sign from Video: using this module user can upload TEST video and then SignNet will extract Hand and Finger features and then input to encoder model to predict TEXT and then predicted TEXT will be input to second decoder model to translate text into SIGN image.

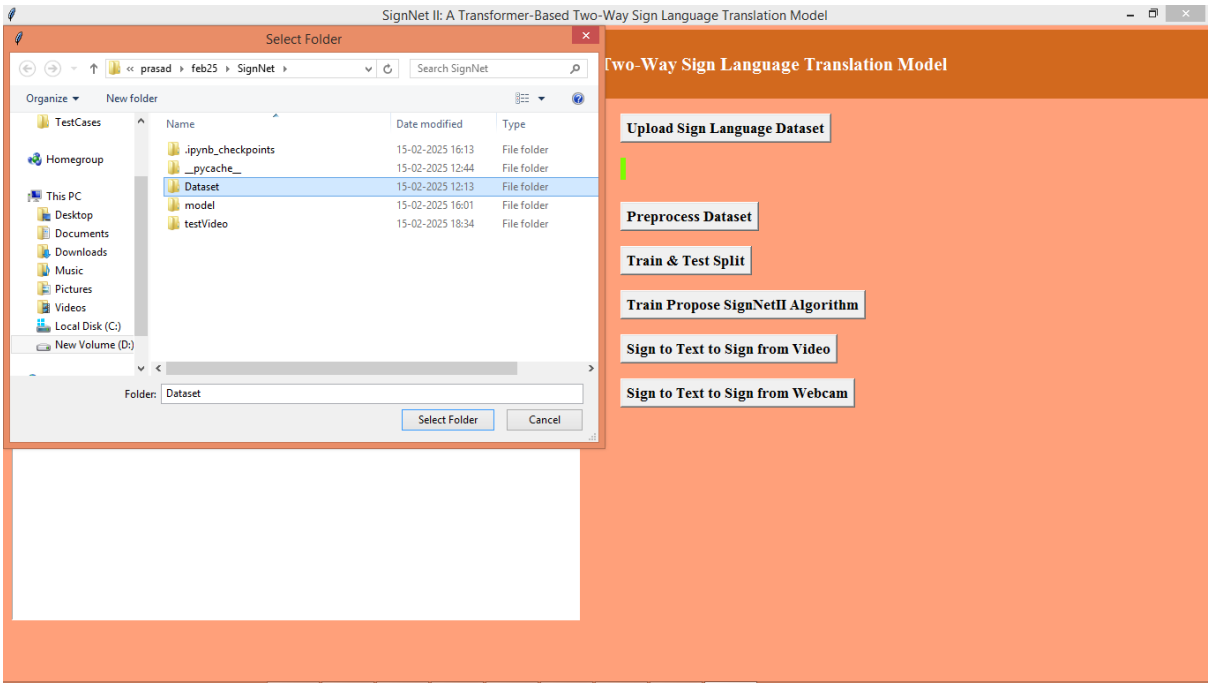
6) Sign to Text to Sign from Webcam: same sign to text and text to sign can be predicted using LIVE webcam. Here you need to show exact hand sign as per dataset to make correct prediction.

SCREEN SHOTS

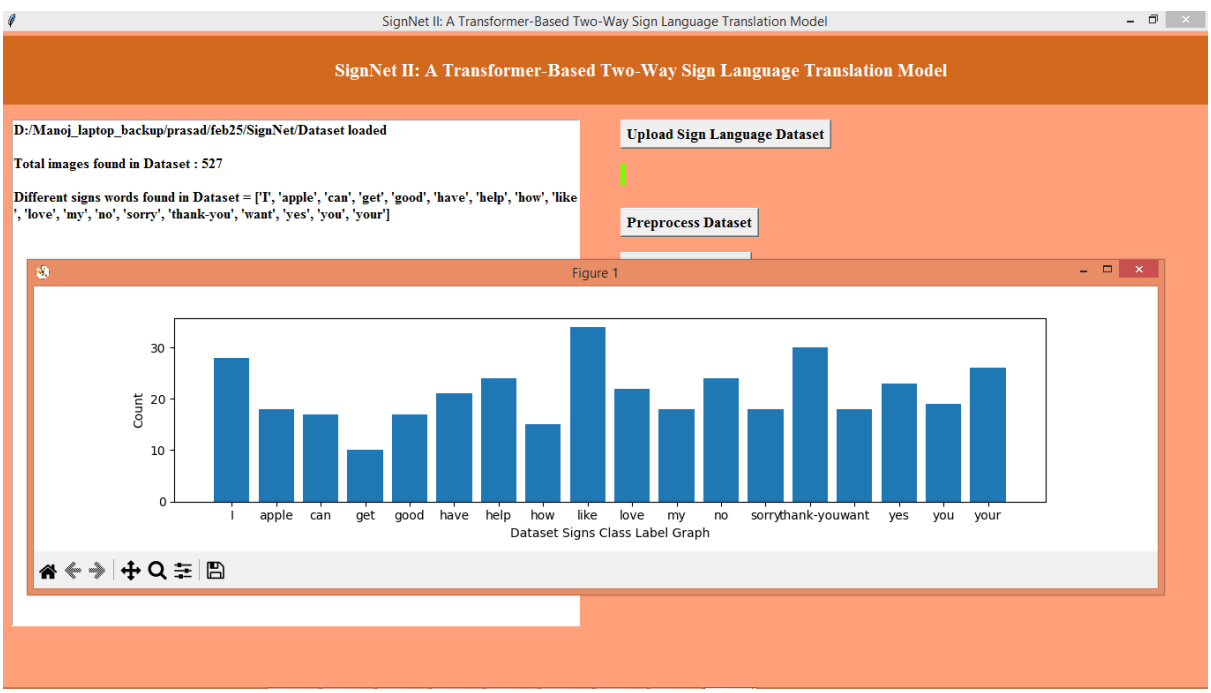
To run project double click on 'run.bat' file to get below page



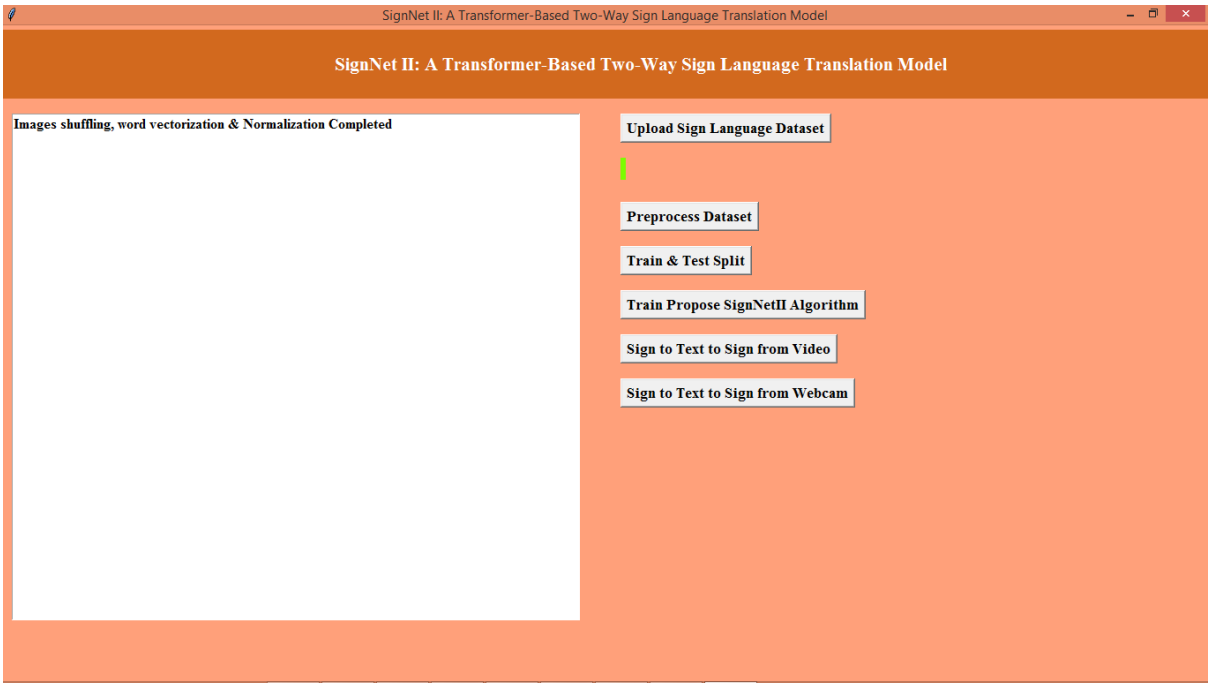
In above screen click on ‘Upload Sign Language Dataset’ button to get below page



In above screen selecting and uploading entire ‘Dataset’ folder and then click on ‘Select Folder’ button to load dataset and then will get below page



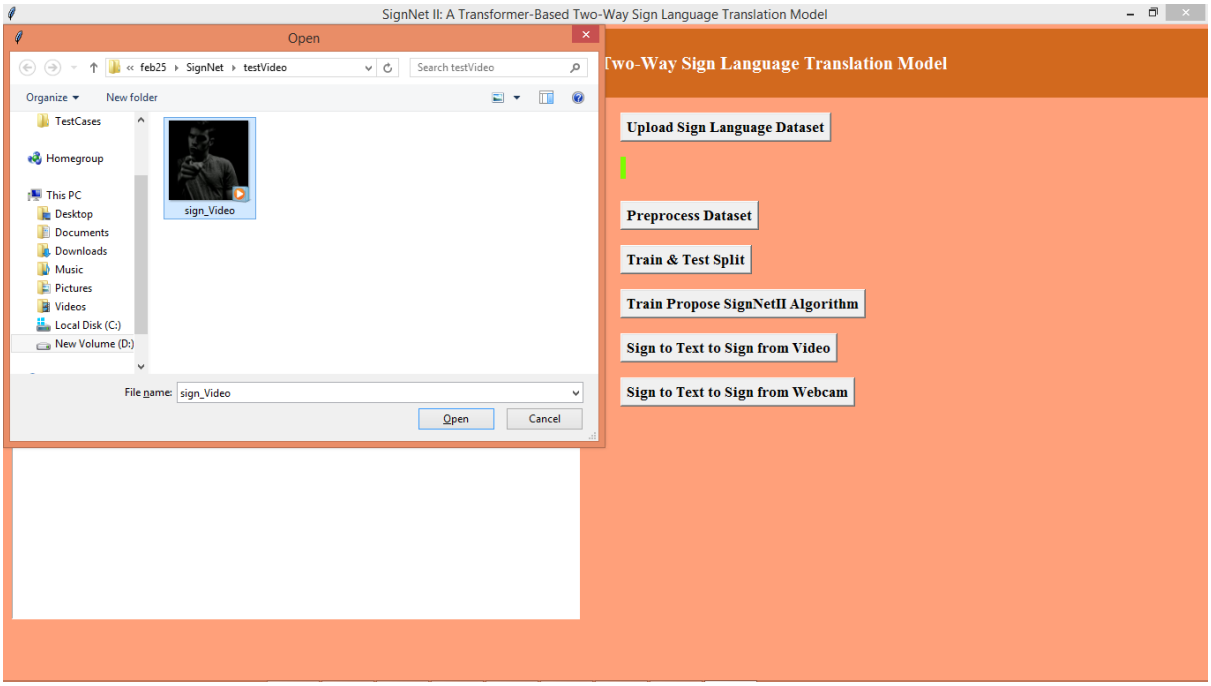
In above screen in text output can see dataset contains 527 images from 18 different class label words and in graph x-axis represents ‘WORD’ name and y-axis represents number of images available for that word. Now click on ‘Pre-process Dataset’ button to clean and normalize dataset values and then will get below page



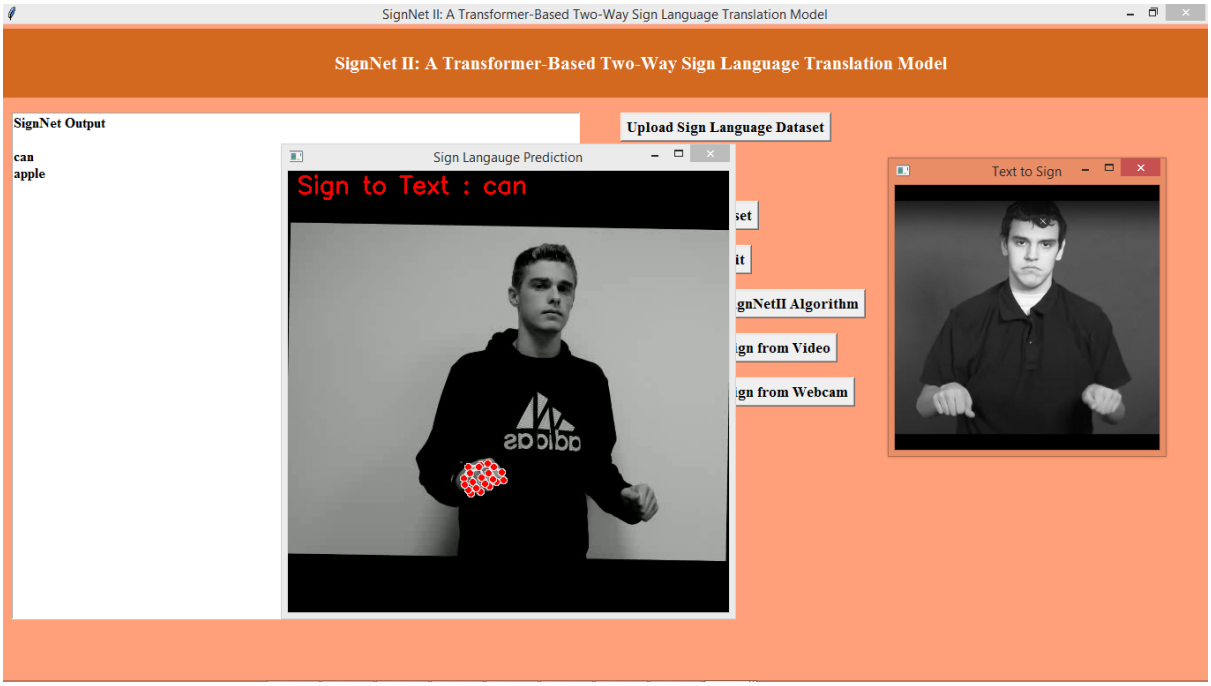
In above screen dataset normalization completed and now click on ‘Train & Test Split’ button to split dataset and then will get below page

In above screen can see train and test size data and now click on 'Train Propose SignNetII Algorithm' button to train a model and then will get below output

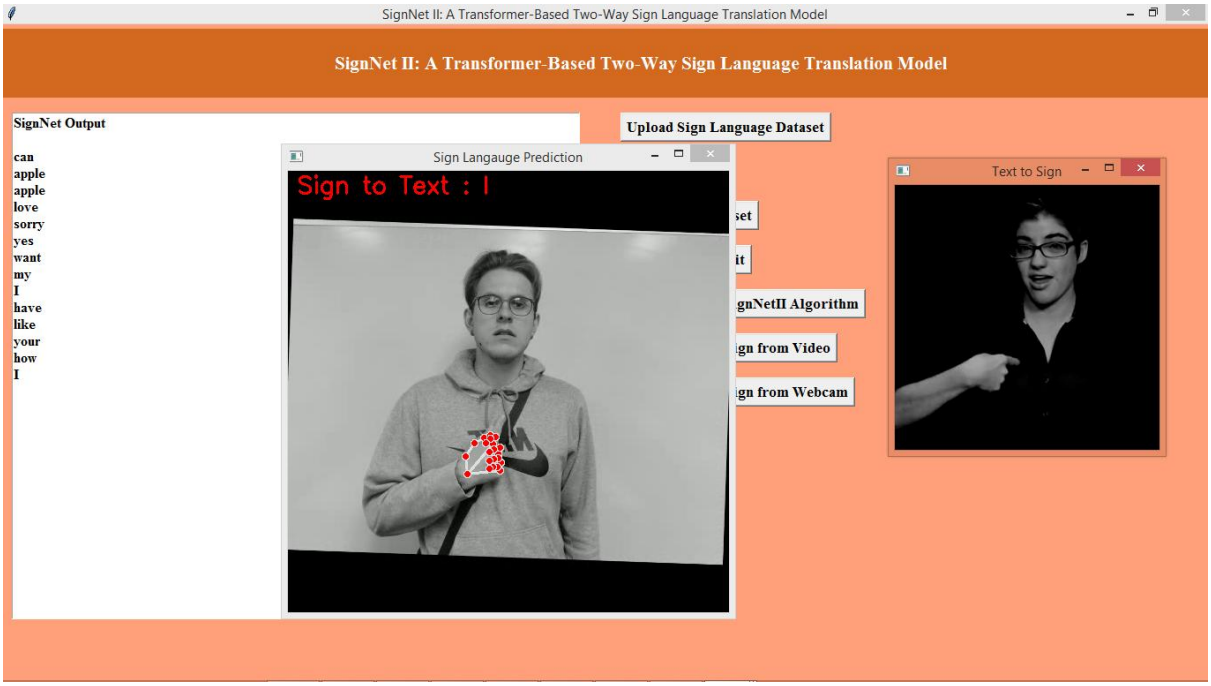
In above screen propose SignNet got 0.5% BLEU score on predicted words from given TEST data and now click on ‘Sign to Text to Sign from Video’ button to upload test video and then will get below output



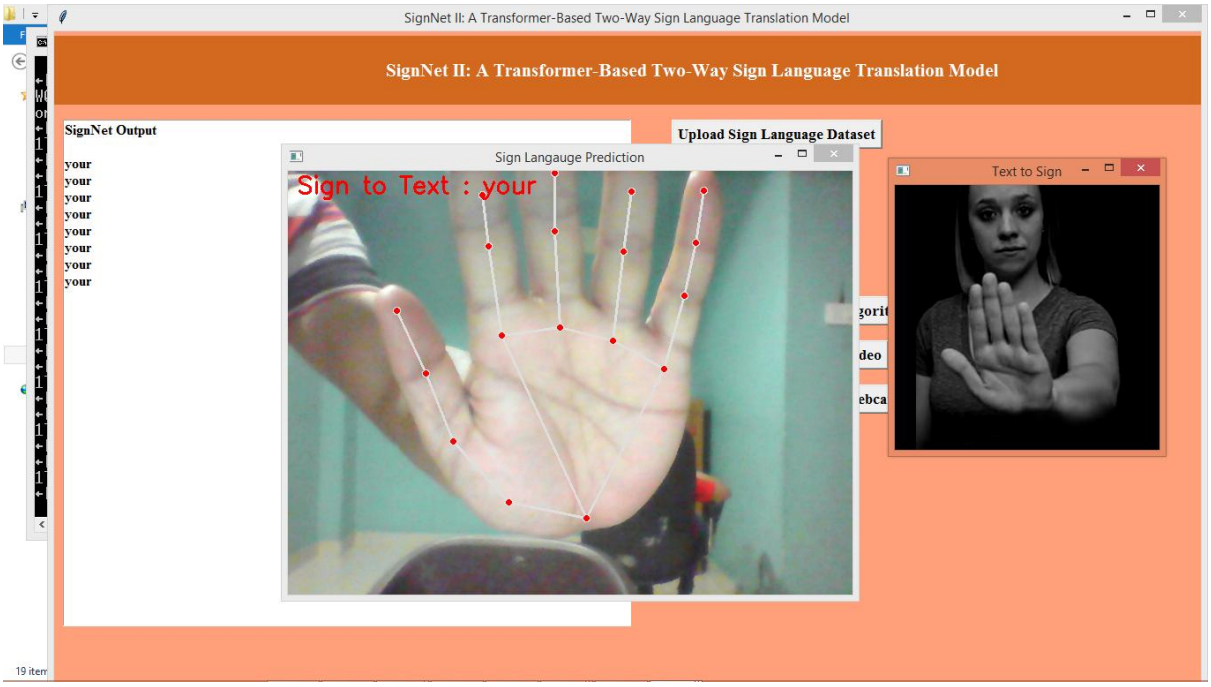
In above screen selecting and uploading test video and then press ‘open’ button to get below output



In above screen on hand we can see red dots which are nothing but open pose features extraction and then in red text can see predicted SIGN TEXT as ‘can’ and then second image is the generated or predicted image from predicted TEXT. Below is another frame output



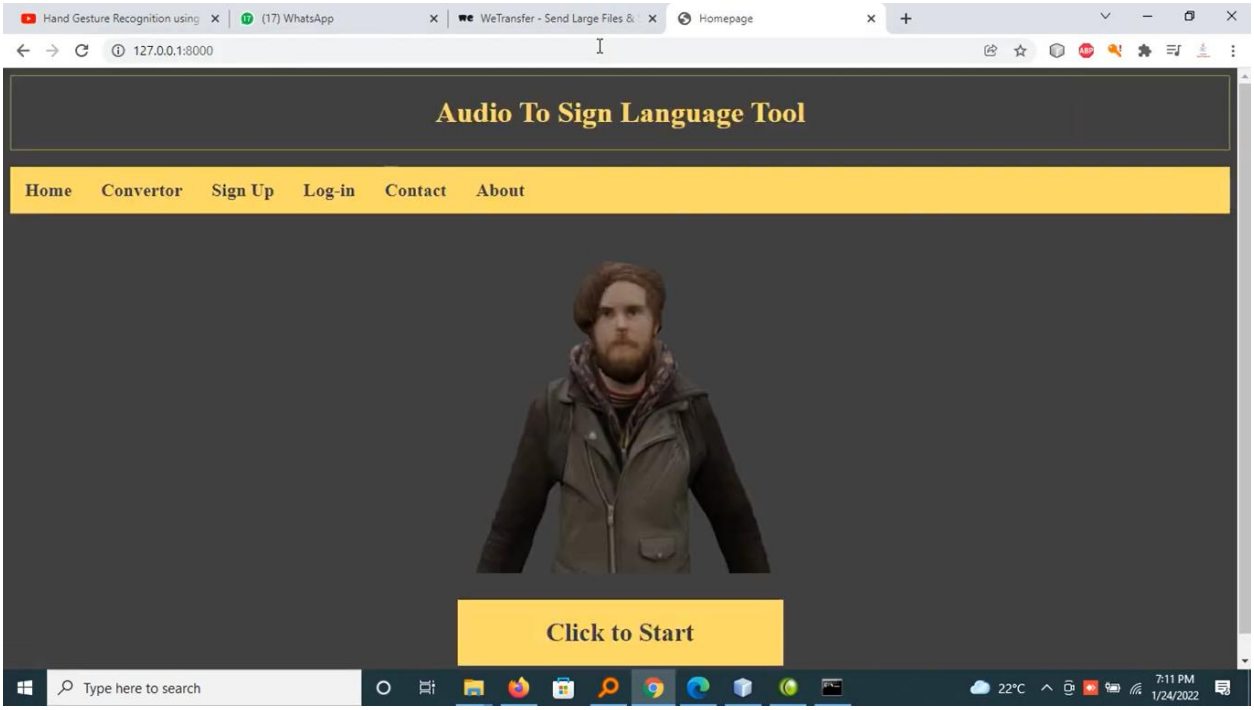
In above screen predicted word is ‘I’ and similar SIGN image also generated which can see in second image. Now click on ‘Sign to Text to Sign from Webcam’ button to start webcam and then show you hand sign to web cam to get below output



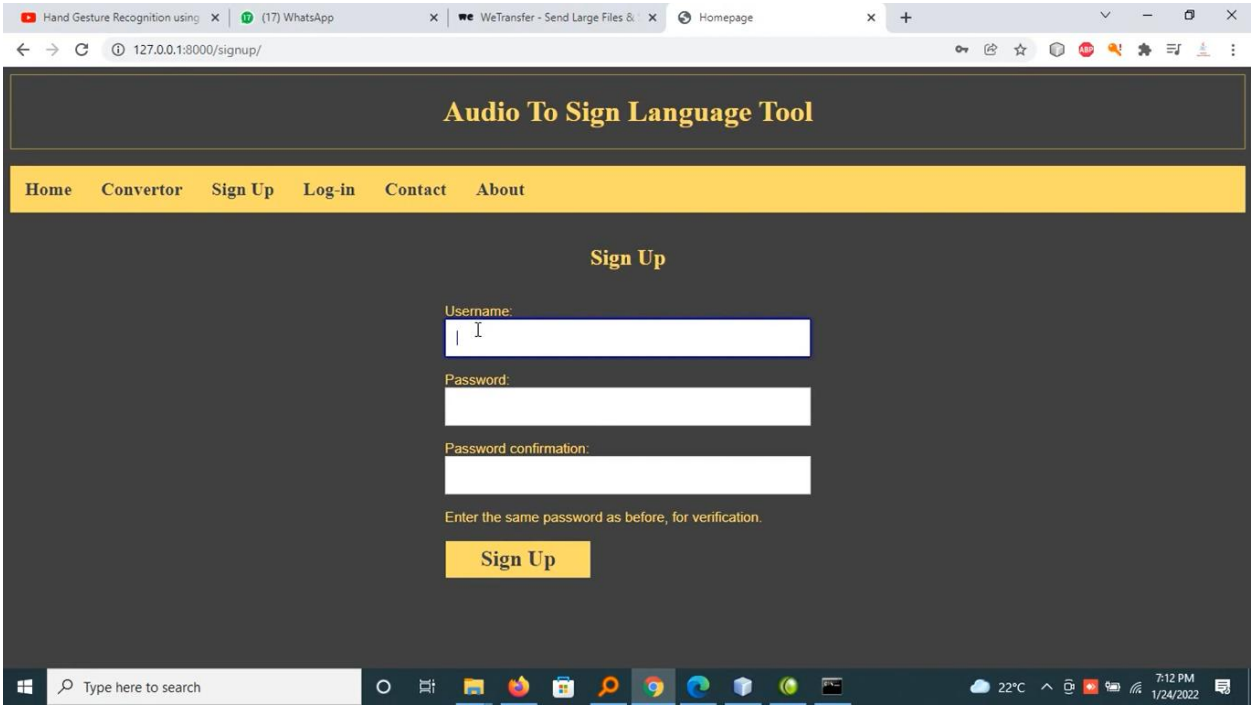
In above screen i am showing some sign from hand and this predicted as ‘your’ and same TEXT is converted to image which can see in second image.

Text to Sign:

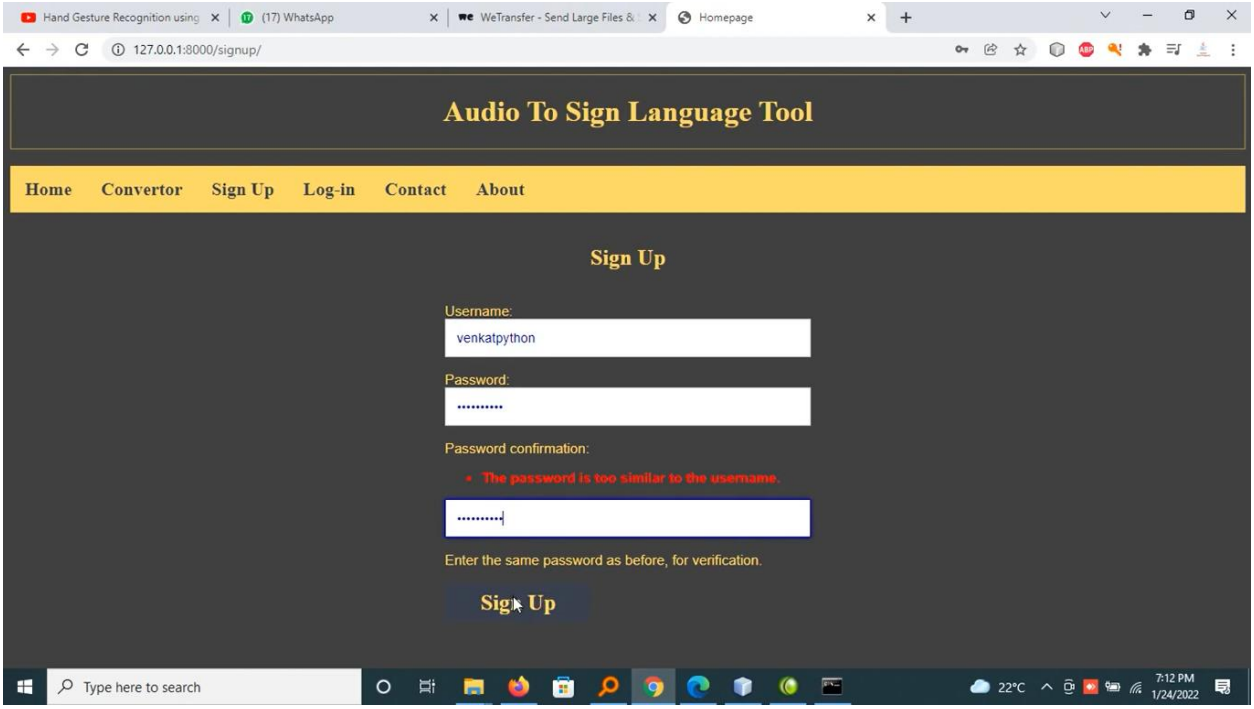
1.Homepage



2.User Registration



3.


Hand Gesture Recognition using
(17) WhatsApp
WeTransfer - Send Large Files &
Homepage
127.0.0.1:8000/signup/
Audio To Sign Language Tool
Home
Convertor
Sign Up
Log-in
Contact
About
Sign Up
Username:
venkatpython
Password:
Password confirmation:
The password is too similar to the username.
Enter the same password as before, for verification.
Sign Up
Type here to search
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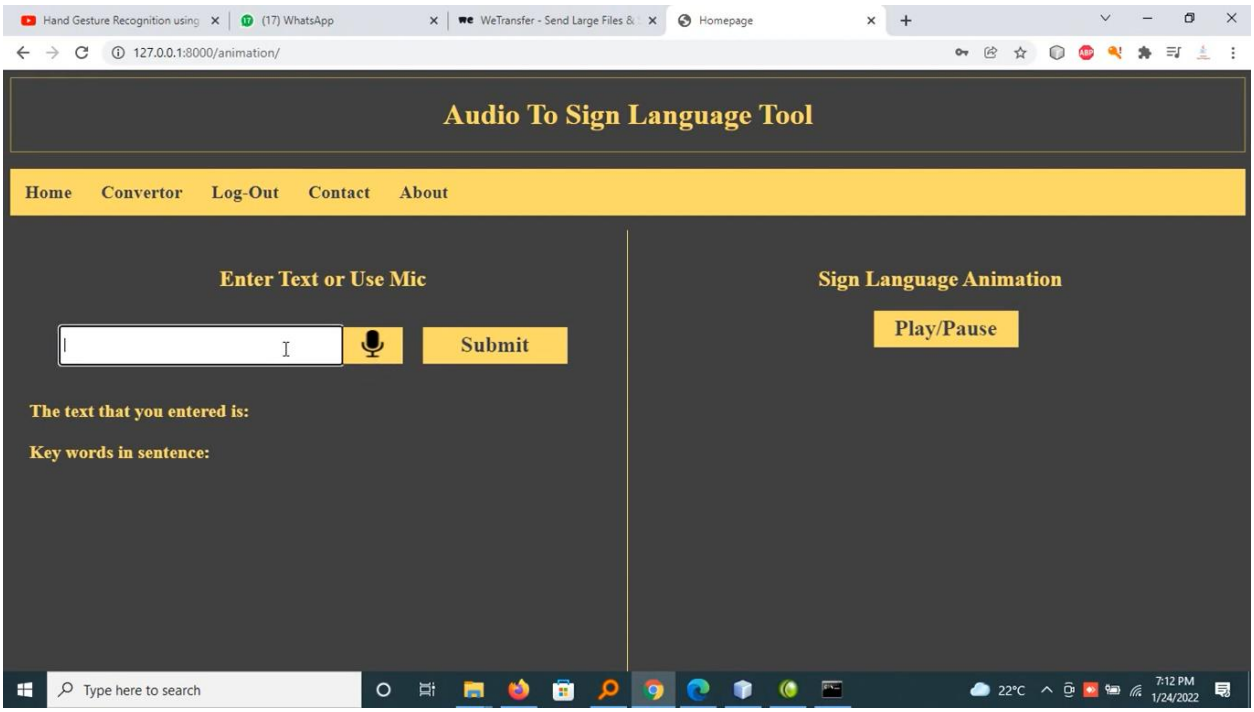
1/24/2022

4.


Hand Gesture Recognition using
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WeTransfer - Send Large Files &
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Audio To Sign Language Tool
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Enter Text or Use Mic
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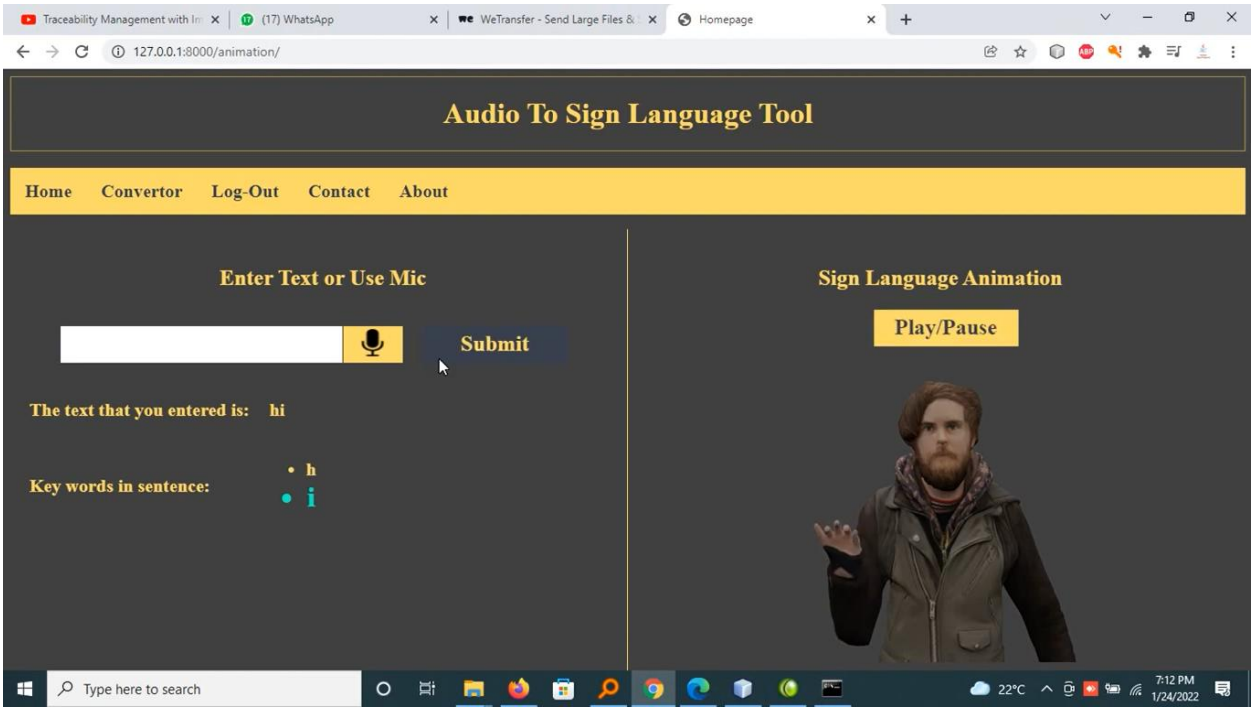
5.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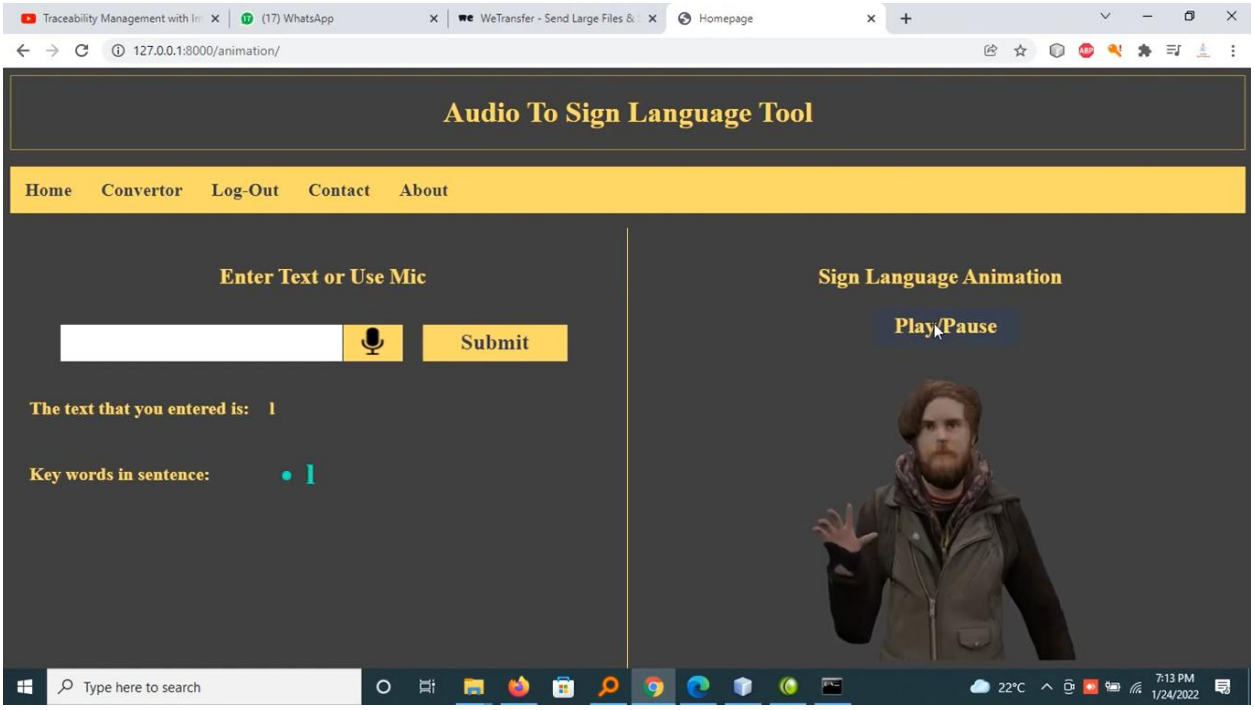
6.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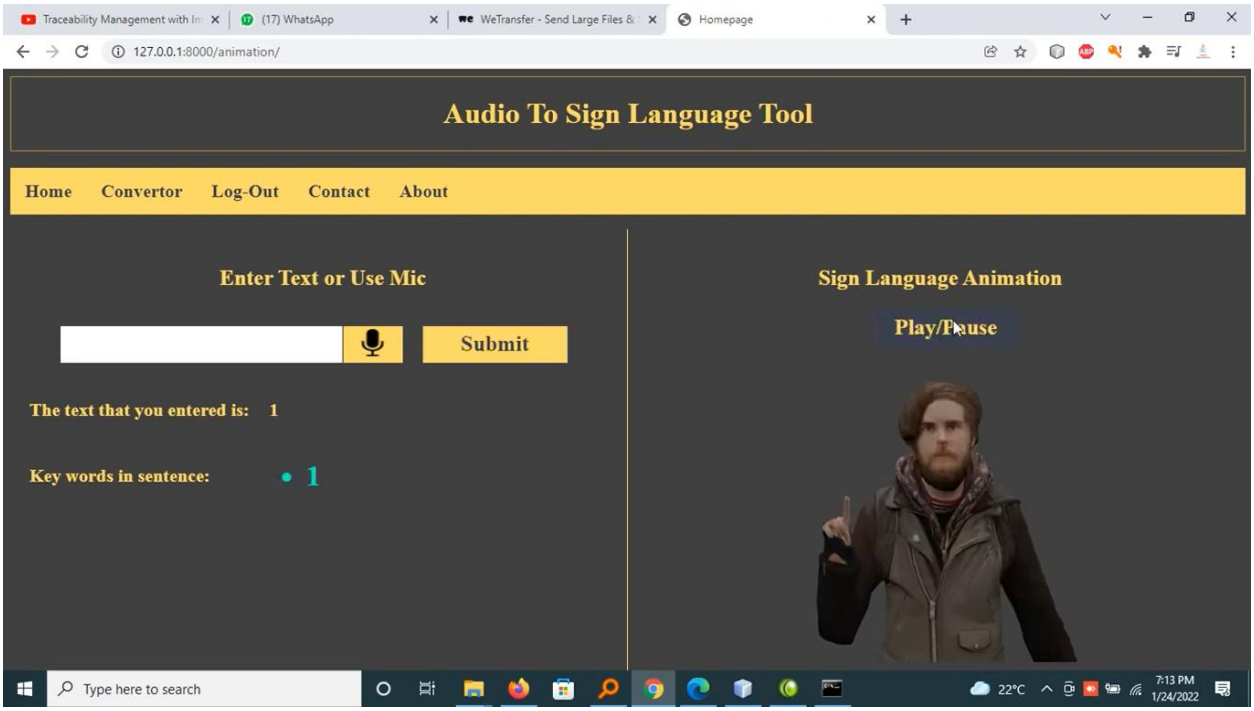
Traceability Management with In
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Sign Language Animation
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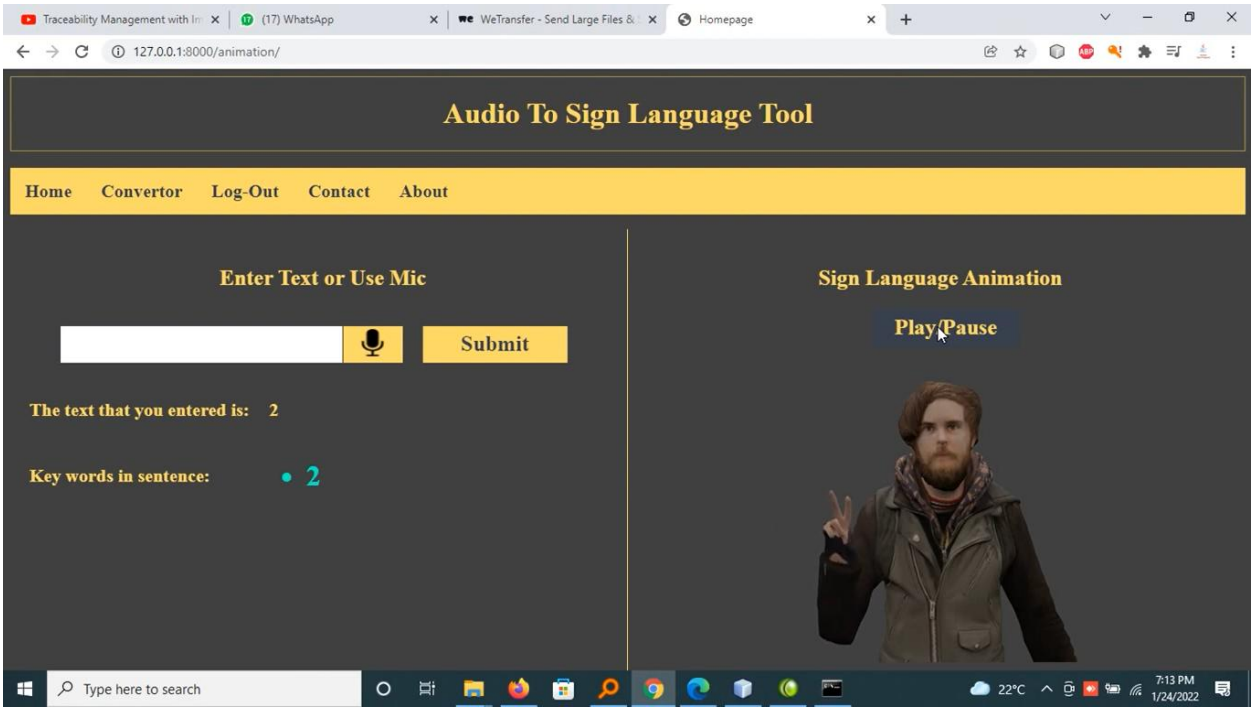
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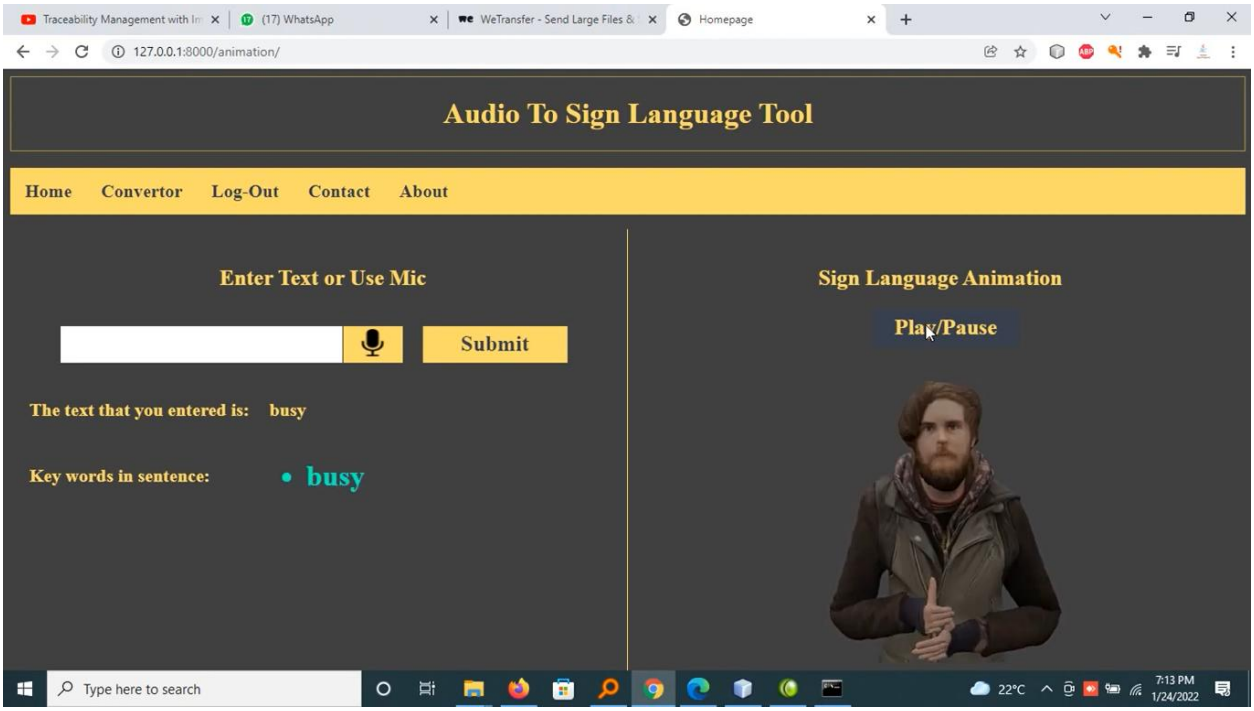
Traceability Management with In
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Audio To Sign Language Tool
Home
Convertor
Log-Out
Contact
About
Enter Text or Use Mic
Submit
The text that you entered is: busy
Key words in sentence:
busy
Sign Language Animation
Play/Pause
Type here to search
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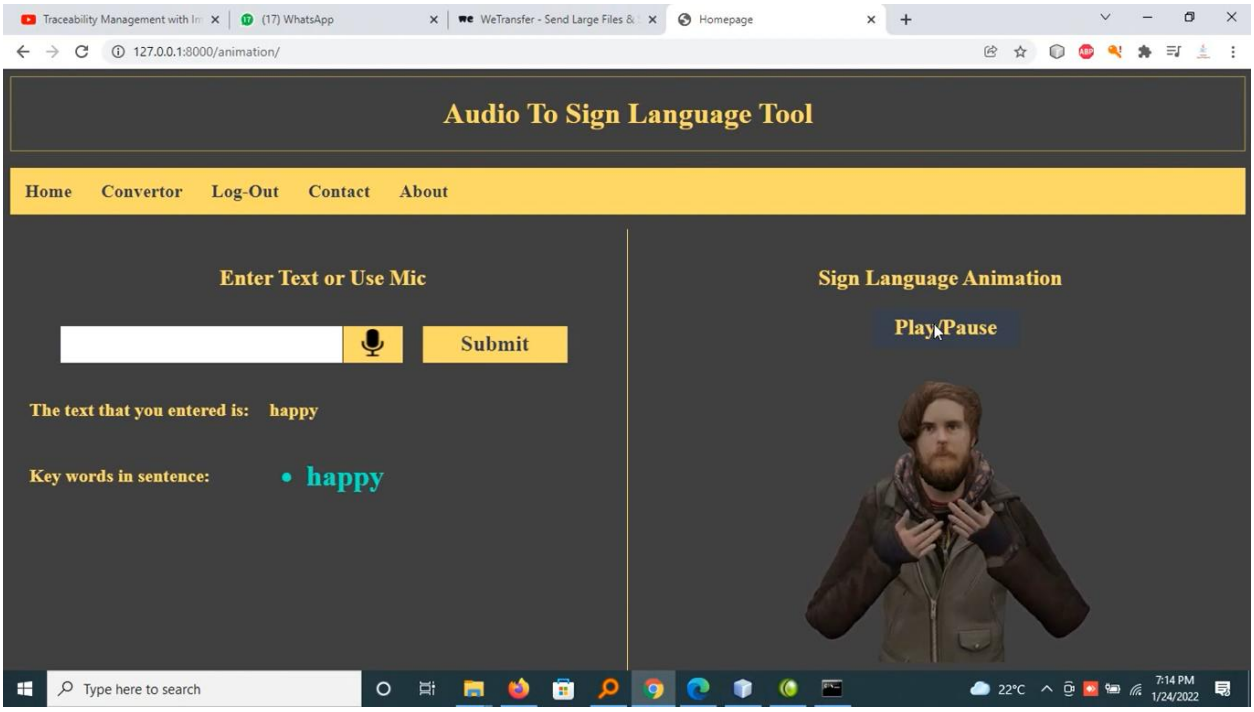
1/24/2022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Audio To Sign Language Tool
Home
Convertor
Log-Out
Contact
About
Enter Text or Use Mic
Submit
The text that you entered is: happy
Key words in sentence:
happy
Sign Language Animation
Play/Pause
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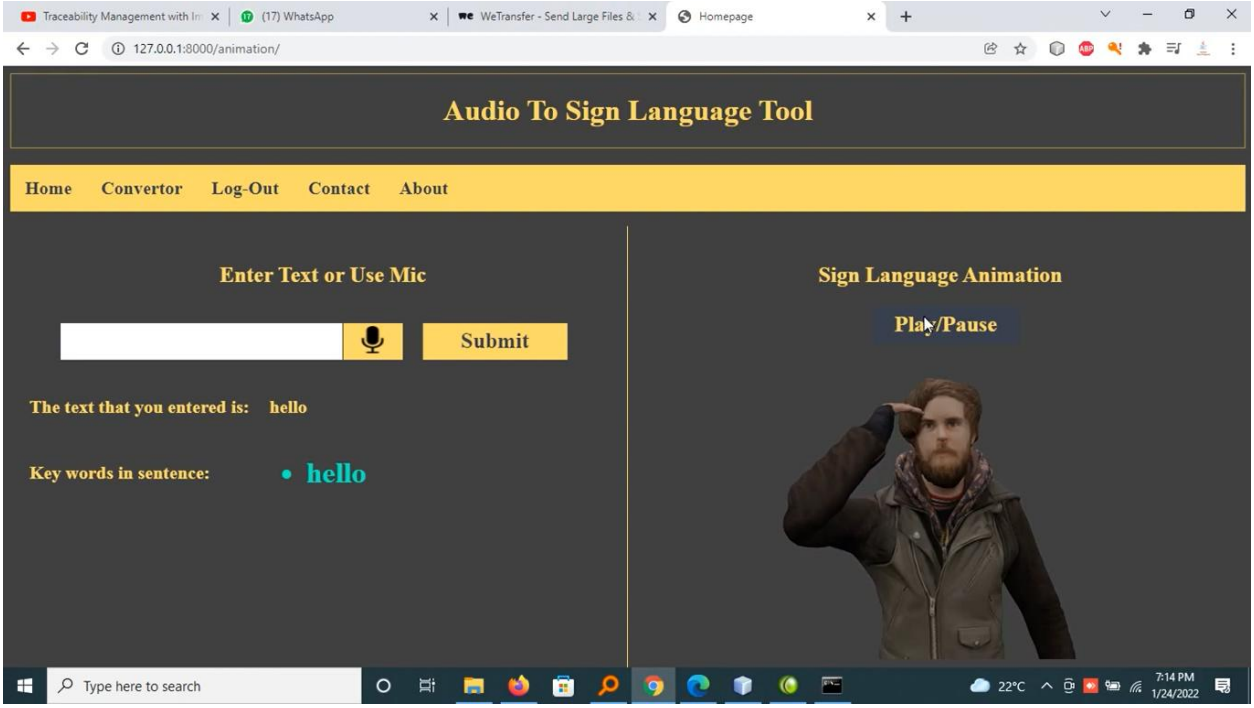

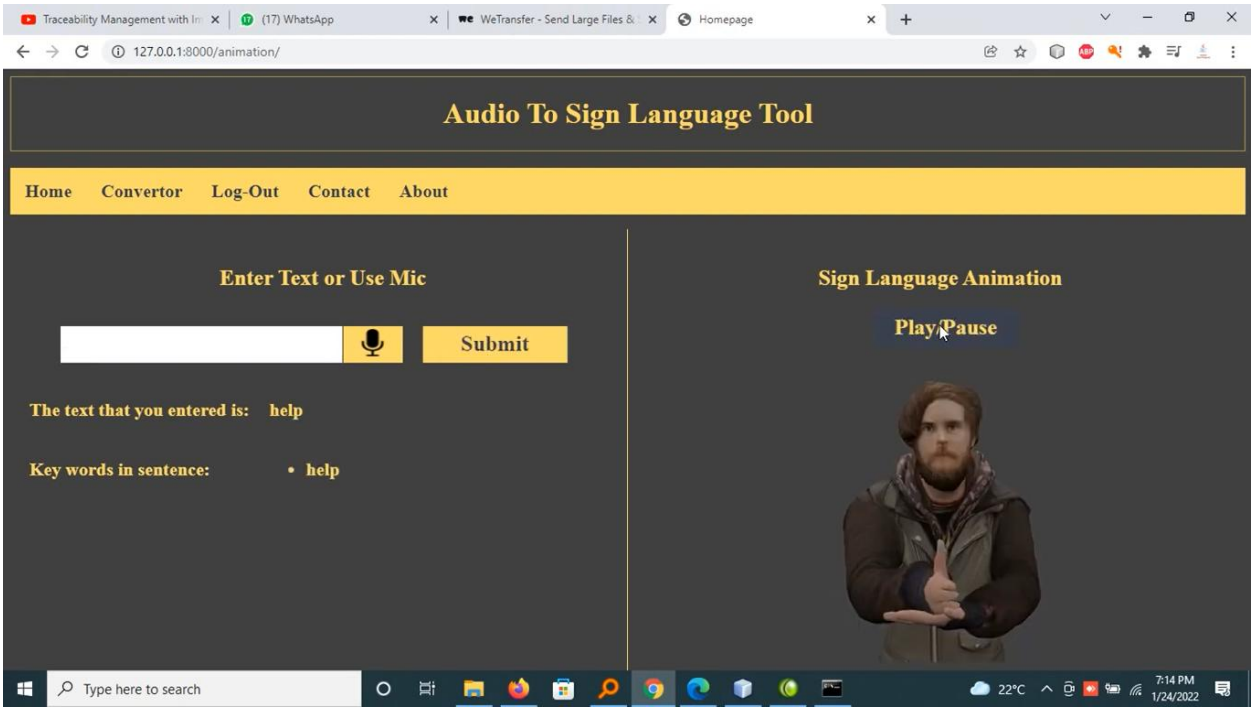

Similarly by following above screens you can run code with both video and webcam.

Note: this project is very heavy in training and prediction as its using multiple models to extract features and for prediction. So we implemented necessary model. While prediction don't run continuous as there is risk of your hard-disk or SSD crash.

Note: from webcam to get correct output your hand sign must be accurate as per dataset.

Conclusion:

In this paper, we presented **RIFD-NET**, a robust and efficient image forgery detection network that integrates spatial and frequency domain features with attention mechanisms to accurately identify and localize manipulated regions. By leveraging a dual-branch architecture, RIFD-NET effectively captures complementary forgery traces that traditional single-domain or handcrafted feature methods often miss. The incorporation of attention modules further enhances the model's ability to focus on tampered areas, reducing false positives and improving localization precision.

Extensive experiments on benchmark datasets demonstrated that RIFD-NET outperforms existing state-of-the-art forgery detection approaches in terms of accuracy, robustness, and generalization across diverse forgery types and image conditions. Additionally, the system's computational efficiency enables practical deployment in real-world scenarios, making it suitable for applications ranging from digital forensics to social media content verification.

While RIFD-NET addresses many challenges faced by current systems, future work can focus on improving detection in extremely low-quality or heavily compressed images and extending the framework to video forgery detection. Overall, RIFD-NET

represents a significant step forward in enhancing the reliability and trustworthiness of digital images in an era where image manipulation is increasingly prevalent.

Future Work:

While RIFD-NET achieves robust performance in detecting various types of image forgeries, several avenues remain for further enhancement and expansion. Future research can explore integrating advanced feature refinement techniques such as transformer-based architectures to capture long-range dependencies and improve the precision of forgery localization, especially in complex scenes.

Another promising direction is extending RIFD-NET to handle video forgery detection, where temporal consistency and motion artifacts present unique challenges. Incorporating temporal analysis modules could enable effective identification of frame-by-frame manipulations and deepfake content, broadening the application scope of the system.

Additionally, enhancing the model's capability to detect forgeries in extremely low-quality or heavily compressed images is critical, as such images often appear in real-world scenarios like social media platforms. Exploring multi-modal data inputs, such as combining image data with metadata or sensor information, may further improve detection accuracy under these challenging conditions.

To support real-time applications, further optimization techniques including model pruning, quantization, and knowledge distillation can be applied to reduce the model's computational requirements without sacrificing performance. Finally, incorporating explainability features into RIFD-NET would provide users with interpretable insights into the detection process, increasing trust and facilitating forensic investigations.

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